

When Silicon Valley Meets Wall Street: A Theory of Financial Overengineering*

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Abstract

We study an equilibrium model in which a financial firm employs an engineer to develop proprietary trading technology for an informational edge. Technological opacity and incentive misalignment generate two self-fulfilling equilibria: a “low-tech” and a “high-tech” equilibrium. In the latter, the engineer overinvests to conform to market beliefs, since otherwise she would be terminated. This equilibrium features lower market liquidity, higher price volatility, and larger trading profits, but also excessive technology costs borne by both parties, rendering it Pareto inefficient. Distinct comparative statics across equilibria yield empirically testable criteria for identifying inefficient technology booms.

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1 Introduction

Financial markets have long been shaped by technological investment, from early computer-based trading in the 1960s (Allen and Gale, 1994; Tufano, 2003) to recent AI-powered trading technologies (IMF, 2024). Each wave of innovation has not only intensified competition among trading firms but also increased demand for engineers and other skilled workers, making technological talent a critical input in financial innovation.¹ Despite this, the existing literature largely focuses on traders’ direct and observable information or speed acquisition as proxies for technology investment, overlooking the distinct role of engineers and the potential incentive misalignment between engineers and trading firms. This gap raises a key question. When the entity that develops trading technology (the engineer) is distinct from the one that deploys it (the trading firm), and they are subject to internal incentive misalignment, how is technology investment determined, and how does it shape financial outcomes such as market liquidity and asset prices?

A central observation motivating our study is the widespread perception that financial technology development is socially excessive. For example, according to a recent survey by Bank of America, a majority of global fund managers believe that companies are spending too much on AI infrastructure.² In addition, Michael Lewis’s *Flash Boys* provides a detailed account of concerns regarding excessive investment in high-frequency trading (HFT). What factors drive “financial overengineering,” defined as excessive investment in financial technology that ultimately harms social welfare? More specifically, how do different forms of technological development, such as internal versus external, and the visibility of technology, namely transparent versus opaque innovation, affect overengineering?

We address these issues by formulating a model, which embeds an incentive problem between the firm and the engineer in developing financial technology, followed by a trading

¹According to *Business Insider*, for example, financial institutions, including banks, hedge funds, and private equity firms, are poaching talent from AI companies amid AI transformations (“AI fever is triggering a new hunt for tech talent on Wall Street,” April 2024).

²See, “Fund managers warn AI investment boom has gone too far,” *Financial Times*, November 2025.

game à la [Kyle \(1985\)](#). In the trading stage, a single informed trader (trading firm) trades a risky asset with a competitive market maker after observing an imperfect signal about the asset’s fundamentals. Rather than acquiring this signal directly, the firm relies on a financial engineer to develop proprietary technology that generates the signal, with higher-level technology providing more accurate information. The engineer improves technology through costly effort, anticipating compensation paid by the firm at the end of the game. Following the classic incomplete contracts literature ([Hart and Moore, 1990](#); [Acemoglu and Pischke, 1999](#)), we assume that the technology level is non-contractible, and the engineer alone bears the non-pecuniary effort cost of development. After the technology is developed, the firm can either terminate or retain the engineer, which translates into a minimum retention requirement on the technology level. The cost asymmetry between the firm and the engineer creates an incentive misalignment: when the problem is severe, the engineer’s individually optimal technology level falls short of the firm’s retention requirement. To avoid termination, the engineer is compelled to develop technology that just satisfies this minimum threshold, meaning that the retention condition is binding. This binding condition is, as we demonstrate below, the key mechanism generating overengineering in our model.

Departing from standard models in which technology or information acquisition is publicly observable, our model allows for technological *opacity*, under which the engineer’s technology development is hidden from the market maker.³ This shift dramatically changes the nature of the incentive problem and gives rise to multiple equilibria: one with lower technology investment (the “low-tech” equilibrium) and another with substantially larger investment (the “high-tech” equilibrium). In the high-tech equilibrium, the trading firm adopts more aggressive trading strategies and improves the price informativeness. However, the market becomes illiquid, and the price is highly volatile. Moreover, all market participants are worse off in the high-tech equilibrium compared to the low-tech equilibrium,

³For example, infrastructure investments in HFT technology, such as building microwave towers or collocation systems, and open-source AI trading strategies (e.g., [Liu, Yang, Gao, and Wang, 2021](#)) are often observable from the outside. In contrast, as illustrated by *Wired* (“[Algorithms Take Control of Wall Street](#),” December 2018), most investments in proprietary trading algorithms are harder to observe externally.

making it Pareto inefficient. This inefficiency arises because a highly precise signal amplifies the firm’s trading profit at the expense of a noise trader, while the costs related to high-level technology ultimately outweigh the gains from trading, reducing the net utilities of both the firm and the engineer.

The key mechanism is the strategic complementarity between the engineer’s incentive to improve the technology level and the market maker’s *belief* about it. When the technology is opaque, the market maker sets the price impact based on a rationally anticipated equilibrium level of technology. Because she is rational, she can compute the equilibrium level, such that the engineer optimally responds by developing exactly the technology level the market maker anticipates. If the market maker anticipates that the high-level technology will be developed, she sets a high price impact to counteract the adverse selection problem (Kyle, 1985). This high price impact would reduce the firm’s trading profit and discourage the firm from retaining the engineer. In turn, the engineer indeed develops high-level technology and boosts trading profits to uphold the firm’s incentive to continue employing the engineer, thereby supporting the high-tech equilibrium. The same logic applies when the market maker believes the technology to be less sophisticated, supporting the coexistence of the low-tech and the high-tech equilibria. We interpret the self-fulfilling nature of multiple equilibria as *fragility* in technology investment.⁴ Once the economy admits multiple equilibria, a mere shift in the market’s belief about unobservable technology can trigger disproportionately large financial technology investments, leading to a highly volatile price, market illiquidity, and a Pareto-inefficient outcome.

Our model shows that multiple equilibria are more likely to emerge when the engineer’s compensation has a large fixed component. A high base salary directly reduces the firm’s share of trading profits and activates the above-mentioned strategic complementarity by tightening the firm’s retention requirement. The engineer, in turn, is willing to overengineer in line with the market maker’s belief: although doing so imposes a substantial effort cost,

⁴As in Greenwood and Thesmar (2011), fragility refers to an economic state that is vulnerable to non-fundamental shifts in model parameters, such as those caused by changes in market beliefs.

deviation would lead to the violation of the retention requirement and the loss of compensation. Large fixed pay makes such deviation costly and motivates the engineer to conform to the market maker’s belief. This mechanism aligns with compensation structures often observed in the finance industry, where high base salaries and signing bonuses are common.⁵ Our model thus provides a consistent link between such wage structures and inefficient overengineering.

Comparative statics differ markedly across the high-tech and low-tech equilibria. When the firm’s surplus changes due to an exogenous shock, the engineer adjusts technology development to restore the firm’s retention incentive. In the high-tech equilibrium, where investment is already excessive, the firm’s marginal utility is negative, and holding back from further overinvestment helps recover the firm’s utility, while the opposing mechanism applies to the low-tech equilibrium. This asymmetry causes the engineer’s innovation to respond differently across the two equilibria, generating two key implications. First, since technology influences the asset’s price and financial market quality, observing market responses to exogenous shocks can empirically distinguish an inefficient high-tech equilibrium. While prior literature, particularly in the context of high-frequency trading (e.g., [Budish, Cramton, and Shim, 2015](#)), has raised concerns about socially wasteful innovation, our model provides a framework to empirically separate inefficient technology investments. Second, the multiple equilibria offer a theoretical rationale for the mixed empirical evidence surrounding the effects of contractual or labor market interventions on innovation, such as non-compete clauses (e.g., [Werner, 2023](#); [Lee, 2024](#)). In our framework, seemingly similar interventions can have opposing effects on innovations and financial markets depending on whether the economy is in the high-tech or low-tech equilibrium.

We consider two extensions of the baseline model. In the first, we introduce competition among trading firms and show that as competition compresses individual trading profits,

⁵Large fixed compensation in the financial sector is generally supported in the literature. Empirically, [Philippon and Reshef \(2012\)](#) document a large finance wage premium, while [Glode and Lowery \(2016\)](#) and [Acharya, Pagano, and Volpin \(2016\)](#) provide theoretical analyses.

retention requirements tighten and the strategic complementarity becomes more likely to activate, making overengineering more prone to arise under oligopolistic market structures. In the second extension, we endogenize the opacity of financial technology by allowing the trading firm to choose between transparent and opaque technology, capturing the real-world practice where trading firms often strategically obscure their innovations.⁶ We show that the firm faces a tradeoff: while opacity encourages technology development through an inelastic price impact, it potentially triggers the Pareto-inefficient high-tech equilibrium, providing a unified explanation for both transparent and opaque innovations.

While our benchmark discussion focuses on an engineer developing proprietary trading technology, the core mechanism extends beyond this specific mode of trading-edge acquisition. More generally, our model captures the internal generation of firm-specific information through human capital. In contrast to third-party information or technology vendors, who operate as monopolists by serving multiple clients (as described in [Admati and Pfleiderer, 1986](#) and subsequent studies), in our setup the trading edge is developed by an entity other than the firm (i.e., the engineer or analyst) whose compensation includes a fixed component. This arrangement is consistent with settings where trading-relevant human capital is firm-specific and cannot be readily redeployed across firms due to contractual or legal frictions, such as NDAs or non-compete clauses. Our results show that this mode of trading-edge acquisition, in turn, sows the seeds of overinvestment in trading edges.

Our study is closely related to the literature on information and speed acquisition in financial markets. Traditional models, such as [Grossman and Stiglitz \(1980\)](#) on information acquisition, and more recent works, such as [Foucault et al. \(2013\)](#), [Foucault, Kozhan, and Tham \(2017\)](#), and [Huang and Yueshen \(2021\)](#) on speed acquisition, focus on traders' incentives while abstracting away from the role of entities creating these advantages. The

⁶Legal disputes over proprietary trading algorithms highlight such an incentive, as reported by *The Wall Street Journal* ("[Legal Suit Sheds Light on Secret Trading Technology](#)," June 2015). Also, many high-frequency trading firms try to hide their technology purchases from rivals, suggesting the importance of opacity in their technology investments (*The Wall Street Journal*: "[Trading Tech Accelerates Toward Speed of Light](#)," August 2016).

issue of overinvestment in financial technology has been understood as a result of a prisoner’s dilemma among trading firms, featuring strategic substitution (Glode, Green, and Lowery, 2012; Biais, Foucault, and Moinas, 2015; Budish, Cramton, and Shim, 2015). A few studies have instead derived excessive information acquisition as one of multiple equilibria arising from strategic complementarity.⁷ Fishman and Parker (2015) and Bolton, Santos, and Scheinkman (2016) show that information acquisition enables cream-skimming, which worsens the lemons problem faced by uninformed traders and thereby raises the rent to becoming informed, generating positive feedback.⁸ In these papers, complementarity arises from externality across competitive traders in fragmented markets. Our model offers an entirely different, more micro-level mechanism, in which an internal incentive problem interacts with technological opacity. This distinction yields several implications unique to our framework. For example, our multiple equilibria are strictly Pareto-ranked, and every party in the model is worse off in the high-tech equilibrium, while inefficiency in the above studies is defined in terms of aggregate welfare. Moreover, since the inefficiency in our model stems from an internal incentive problem within the firm rather than from an externality among traders, addressing it calls for policies that target internal incentives and technological opacity rather than competition. Also, in our model, observably similar firms trading in different markets can settle into different equilibria; the model provides implications for engineer compensation as well.

The literature following Admati and Pfleiderer (1986, 1988, 1990) introduces an information seller but typically assumes a monopolistic seller who is endowed with information and extracts the full traders’ surplus by offering take-it-or-leave-it contracts. Veldkamp (2006) describes frenzies in a perfectly competitive information market driven by decreasing average

⁷Biais, Foucault, and Moinas (2015), cited above for their substitution result, also obtain complementarity only in a special case that hinges on a particular structure of traders’ heterogeneous private valuations; under more natural parameter assumptions, their own Corollary 1 shows that strategic substitution arises instead.

⁸Kurlat (2019) studies the wedge between the private and social value of financial expertise. Using a model with a unique competitive equilibrium, he empirically quantifies the wedge for a specific market, rather than analyzing conditions under which multiple equilibria with overinvestment arise.

costs of information production. These models, however, highlight the non-rival nature of informational services, a feature that does not extend to financial technologies or tech workers intended for exclusive use by a single trading firm. Our model explicitly incorporates incentive misalignment and explores overinvestment in such trading edges when the firm acquires them through firm-specific human capital.

The rest of the paper proceeds as follows. Section 2 presents our model framework, and Section 3 analyzes its equilibrium, showing the possibility and implications of overengineering. Section 4 examines key ingredients of our main result, and Section 5 considers extensions. Section 6 provides discussion. The Appendix contains all proofs of the theoretical results.

2 Model

Consider a two-period model with four risk-neutral agents: a trading firm, an engineer, a market maker, and a noise trader.⁹ In the first period, the trading firm has an engineer on staff, and she develops proprietary technology for trading.¹⁰ In the second period, the trading firm leverages the technology and participates in the financial market, where a single risky asset is traded. The asset's payoff, δ , is realized at the end of the game and follows a normal distribution with mean $\bar{\delta}$ and variance σ_δ^2 .

2.1 Technology and Compensation

Using the technology developed by the engineer, the firm obtains a noisy signal about the asset's payoff, $s = \delta + \epsilon$, where the noise term, ϵ , is normally distributed with mean 0 and variance σ_ϵ^2 . We define the level of technology by the reliability ratio of the signal:

⁹Following the convention, we assume that competitive market makers exist, while only one of them actively executes incoming orders on the equilibrium path. It ensures the competitive price, as other market makers would undercut any non-competitive prices off the equilibrium path.

¹⁰Our model features the expertise asymmetry between a financial firm and a technology specialist (Aghion and Tirole, 1997): while the firm employs the engineer, the engineer retains decision rights over the technological specifications. For example, the firm can identify trading needs (such as faster execution) but lacks the technical knowledge to specify the optimal level of technology.

$\varphi \equiv \text{SD}[\delta]/\text{SD}[s] = \sqrt{\sigma_\delta^2/(\sigma_\delta^2 + \sigma_\epsilon^2)}$, where $\varphi = 0$ corresponds to the lowest technology level and $\varphi = 1$ to the highest. It measures how much uncertainty in δ is resolved by observing the signal and is directly related to the signal precision, σ_ϵ^{-2} , and the signal-to-noise ratio.¹¹

To motivate the engineer's technology development, the firm pays a performance-based affine wage, $w = a + b\Pi$, to the engineer after the realization of the net trading profit Π (formalized below), where $a \geq 0$ and $b \in [0, 1]$ are exogenous parameters capturing, respectively, the engineer's fixed pay and the performance sensitivity of her compensation. We take this wage structure as given in the baseline model and characterize the equilibrium in terms of (a, b) ; Section 6 provides a microfoundation, where this structure and the parameters are endogenously determined by the bilateral relationship between the firm and the engineer.¹²

The engineer controls $\varphi \in [0, 1]$ by incurring non-pecuniary *development cost* $C_e(\varphi) = c_e\varphi^2$ with $c_e > 0$, which we interpret as disutility or the opportunity cost of time and effort required for technology development.¹³ As we specify below, technology is opaque, meaning that φ is not observable (verifiable) from outside. Hence, φ itself is not directly contractible, and the firm retains the option to terminate the engineer upon observing the realized φ .¹⁴ Specifically, if the expected profit from trading on the technology is insufficient to cover the committed wage, the firm dismisses the engineer, leaves the technology unused, and, due to the lack of informational advantage, stays inactive in the subsequent trading stage. Anticipating this termination possibility, the engineer chooses φ to maximize her expected wage net of the development cost.

¹¹The engineer actually controls σ_ϵ^2 , while it corresponds one-to-one with determining φ . Hence, for notational simplicity, we henceforth assume that she controls φ .

¹²Section 6 establishes that our baseline setting is equivalent to an environment in which the firm can terminate the engineer after technology development and recover the technology with some probability at a fixed salvage cost.

¹³Alternatively, we may assume that the engineer determines the maximum level of technology, φ_{max} , and the firm adjusts its utilization rate following $\varphi \leq \varphi_{max}$. With this setting, the equilibrium outcome remains unchanged, as the firm fully utilizes the technology. See Subsection 4.2 for the results when the firm directly chooses φ .

¹⁴This contractual friction is consistent with the labor economics literature (e.g., Acemoglu and Pischke, 1999) that characterizes non-binding wage contracts. It is supported by the fact that the technology level is difficult to verify from an outsider's perspective (e.g., a court), in line with our objective of analyzing opacity in financial technology.

2.2 Technology Opacity

Throughout the model, we assume that the level of technology φ is observable to the trading firm through its direct communications with the engineer. However, technology *opacity* prevents the market maker from observing φ . Nevertheless, the market maker rationally computes and anticipates the equilibrium technology level φ^* , as it is a deterministic and constant technology level that the engineer would choose. We assume that φ^* is common knowledge and search for equilibria in which the engineer actually chooses $\varphi = \varphi^*$ as her optimal response to the market's belief φ^* , while no restrictions are imposed on the engineer's choice off the equilibrium path, meaning that φ may potentially deviate from φ^* . Importantly, technology opacity prevents the firm and the engineer from affecting the market maker's belief and thereby influencing the pricing strategy.¹⁵

2.3 Financial Market

In the process of extracting the signal and trading on it, the firm incurs the *maintenance cost* of the technology, $C_f(\varphi) = c_f \varphi^2$ with $c_f > 0$. It could arise from the costs of applying the technology to practical market situations and the expenses of maintaining or updating the equipment. The cost is increasing in the technology level φ , reflecting the fact that more sophisticated technologies require a higher maintenance cost.¹⁶

The trading stage is based on Kyle (1985), where the risky asset is traded among the firm, the market maker, and the noise trader. The trading firm places a market order for $x \in \mathbb{R}$ units of the asset and earns the trading profit $\pi = (\delta - p)x$ in the end of the trading stage. The price of the asset p is set by the market maker upon observing the aggregate

¹⁵The firm and the engineer do not have a commitment device and cannot convey signals about φ to the market maker in a credible manner. Even if the engineer deviates from the equilibrium technology level, which may happen off the equilibrium path, it does not affect the market maker's computation of φ^* through the noisy order flow, as there always exists a realization of noise trading that makes the order flow consistent with φ^* .

¹⁶Both the technology development and maintenance costs are assumed to be quadratic in φ for tractability, while the main results are robust to more general cost functions, as long as they are weakly convex in φ and satisfy several regularity conditions.

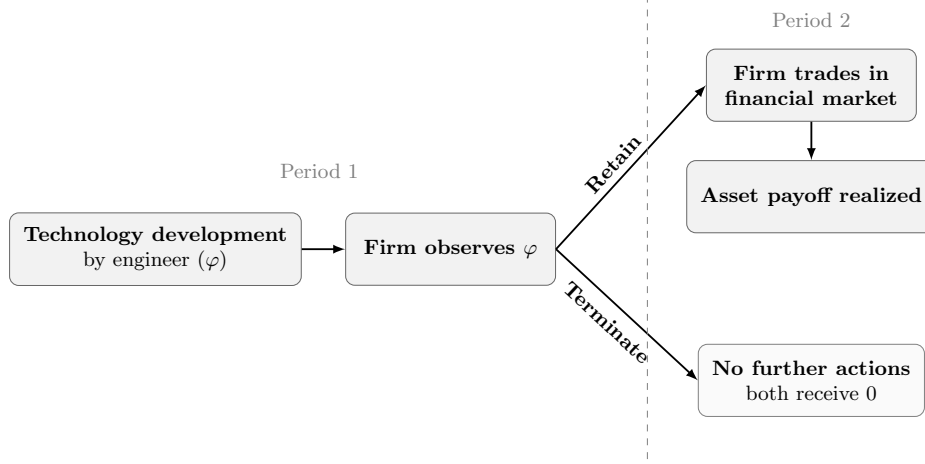


Figure 1: Timing of Events

order flow, $y = x + u$, where $u \sim N(0, \sigma_u^2)$ represents a random market order from the noise trader. As in the standard Kyle (1985) model, the order flow conveys information about δ to the market maker, leading to the semi-strong efficient price:

$$p = E[\delta|y]. \quad (2.1)$$

Upon realization of the asset's payoff, the firm computes the net trading profit, $\Pi \equiv \pi - c_f \varphi^2$, and pays out the compensation to the engineer based on the wage structure specified above.

2.4 Timing and Utility

The timing of the events is summarized in Figure 1, and the model unfolds as follows:

1. (Technology development.) The engineer chooses the technology level $\varphi \in [0, 1]$, incurring the development cost $c_e \varphi^2$.
2. (Retention decision.) Upon completion of technology development, the firm observes φ and decides whether to retain the engineer. If the firm terminates the engineer, it stays inactive in the financial market, and both parties earn zero additional utility. If the firm retains the engineer, it derives the noisy signal s by incurring the maintenance cost $c_f \varphi^2$ and moves on to the trading stage.
3. (Trading.) The firm engages in the asset trading based on the observed signal s . In the

end of the trading stage, the asset's payoff δ is realized, and the firm pays compensation w to the engineer.

The firm's net utility is its trading profit minus the maintenance cost and the wage payment to the engineer:

$$U_f = \pi - c_f \varphi^2 - w. \quad (2.2)$$

The engineer's net utility is the wage payment minus the development cost:

$$U_e = w - c_e \varphi^2. \quad (2.3)$$

The market maker earns zero expected profit due to competition, and the noise trader's expected utility (defined in Section 3) mirrors the adverse selection cost imposed by the trading firm.

2.5 Equilibrium

The equilibrium of our model is defined as follows:

Definition 1. *The equilibrium consists of the technology level (φ), the firm's retention decision, the firm's trading strategy (x), and the asset's price (p), such that, (i) the engineer chooses φ to maximize her expected utility, $E[U_e]$, (ii) the firm chooses whether to retain the engineer and selects the trading strategy to maximize its expected utility, $E[U_f]$, (iii) the market maker sets the price according to (2.1), and (iv) every agent's beliefs about other agents' behavior are consistent with their actual behavior.*

In what follows, we impose the following parameter restrictions:

Assumption 2.1. (i) *The base salary satisfies $a < \hat{a} \equiv \frac{(1-b)\sigma_\delta^2\sigma_u^2}{16c_f}$.*

(ii) *The firm's marginal maintenance cost satisfies $c_f > \max\{2c_e, \frac{\sigma_\delta\sigma_u}{2}\}$.*

A substantially high base salary a discourages the firm from retaining the engineer at any technology level, resulting in no equilibrium. The first assumption restricts such situations.

The second condition is to avoid a corner solution ($\varphi = 1$) and to focus on interesting equilibrium scenarios.¹⁷

3 Equilibrium Analysis

Unlike standard models, our setup features both technology opacity and engineer-led technology development. We show that this combination generates multiple equilibria involving overengineering. We demonstrate in Section 4 that each ingredient is necessary.

3.1 Financial Market

The model is solved by backward induction. We conjecture and later verify that the firm's trading strategy in period 2 (the trading stage) takes the following form:

$$x = \beta\varphi^2(s - \bar{\delta}), \quad (3.1)$$

where $\beta > 0$ is an endogenous constant determined later. Namely, the firm's strategy is characterized by the trading intensity, β , multiplied by the firm's informational advantage over the market maker, $\varphi^2(s - \bar{\delta})$.¹⁸

Given the trading strategy (3.1), the semi-strong efficient price in (2.1) is computed by following the linear filtering rule:

$$p = \bar{\delta} + \lambda(\varphi^*)y, \quad (3.2)$$

where

$$\lambda(\varphi^*) = \frac{\beta\varphi^{*2}\sigma_\delta^2}{\beta^2\varphi^{*2}\sigma_\delta^2 + \sigma_u^2}. \quad (3.3)$$

Coefficient λ measures the price impact of order flow. Since the expectation in (2.1) is taken

¹⁷The assumption on c_f is innocuous, and relaxing it yields equilibrium patterns covered as sub-cases in the analyses below.

¹⁸Since $E[\delta|s] = \bar{\delta} + \varphi^2(s - \bar{\delta})$ due to the normal distribution, the difference between the firm's conditional expectation and the market maker's expectation about the asset's payoff is $E[\delta|s] - E[\delta] = \varphi^2(s - \bar{\delta})$.

from the market maker's perspective, it depends on φ^* rather than φ . As a high λ causes a large price reaction to changes in order flow, it represents an illiquid financial market.

Incorporating the market maker's pricing strategy in (3.2), the trading firm maximizes its expected utility, which is equivalent to maximizing expected trading profit, since the maintenance cost is sunk at the trading stage and the affine wage contract does not distort the firm's trading incentive:

$$\max_x E[\pi|s] = (E[\delta|s] - \bar{\delta} - \lambda x) x. \quad (3.4)$$

The conditional expected payoff of the asset is $E[\delta|s] = \bar{\delta} + \varphi^2(s - \bar{\delta})$ and depends on φ rather than φ^* , as the expectation is computed from the firm's point of view. The first-order condition of (3.4) yields the optimal trading strategy as follows:

$$x = \frac{\varphi^2}{2\lambda(\varphi^*)}(s - \bar{\delta}). \quad (3.5)$$

The expression in (3.1) is consistent with (3.5) when

$$\beta = \frac{1}{2\lambda(\varphi^*)}. \quad (3.6)$$

Given φ and φ^* , solving (3.3) and (3.6) yields

$$\beta = \frac{\sigma_u}{\sigma_\delta \varphi^*}, \quad (3.7)$$

and

$$\lambda = \frac{\sigma_\delta}{2\sigma_u} \varphi^*, \quad (3.8)$$

which, together with (3.1) and (3.2), determine x and p . Moreover, the firm's expected

trading profit conditional on its private signal is

$$\mathbb{E}[\pi|s] = \frac{\varphi^4(s - \bar{\delta})^2}{4\lambda(\varphi^*)}. \quad (3.9)$$

The numerator represents the firm's (quadratic) informational advantage over the market maker, while the denominator implies that the firm holds back from trading aggressively when the price impact is high, lowering the expected profit. Similarly, the unconditional expected profit of the firm is given by

$$\bar{\pi} \equiv \mathbb{E}[\pi] = \frac{\sigma_\delta \sigma_u}{2} \frac{\varphi^2}{\varphi^*}. \quad (3.10)$$

Therefore, the firm's technology (φ) directly contributes to the trading profit by expanding its informational advantage, while the technology level believed by the market maker (φ^*) reduces $\bar{\pi}$ by inducing her to increase the price impact.

3.2 Retention Decision

Upon observing φ , the trading firm expects to obtain the following expected utility if it keeps employing the engineer:

$$\begin{aligned} \bar{U}_f &= \mathbb{E}[\pi - w - c_f \varphi^2] \\ &= (1 - b) \left(\frac{\sigma_\delta \sigma_u}{2\varphi^*} - c_f \right) \varphi^2 - a. \end{aligned} \quad (3.11)$$

If, in contrast, the firm decides to terminate the engineer, it receives zero utility. Therefore, the firm is willing to retain the engineer if and only if $\bar{U}_f \geq 0$ holds. We refer to this condition as the *retention condition*. From (3.11), rearranging the condition $\bar{U}_f \geq 0$ yields

$$\varphi \geq R(\varphi^*) \equiv \sqrt{\frac{a}{1 - b} \frac{2\varphi^*}{\sigma_u \sigma_\delta - 2c_f \varphi^*}}. \quad (3.12)$$

This condition imposes the minimum technology level required for retention. Importantly, $R(\varphi^*)$ is an increasing function of φ^* , suggesting that the firm requires a higher technology level for retention when the market maker anticipates a higher equilibrium technology level φ^* . Intuitively, such a belief of high φ^* induces a high price impact, and the firm's expected utility $\bar{U}_f$ decreases, *ceteris paribus* (holding φ fixed). To ensure $\bar{U}_f \geq 0$, the firm therefore tightens its retention condition by raising the minimum required technology level.

3.3 Technology Development

At the technology development stage in period 1, the engineer chooses φ to maximize her expected utility

$$\begin{aligned}\bar{U}_e &= \mathbb{E} [w - c_e \varphi^2] \\ &= a + \left(\frac{b\sigma_\delta\sigma_u}{2\varphi^*} - bc_f - c_e \right) \varphi^2,\end{aligned}\tag{3.13}$$

taking into account the firm's retention condition (3.12) and her own participation condition. If she violates the retention condition, she would not receive the wage payment, and her utility becomes negative due to the technology development cost.

Engineer's marginal utility. Due to technological opacity, the engineer cannot affect the market maker's belief about the equilibrium technology level φ^* and therefore cannot internalize the adverse effect of a stronger price impact on her utility. As a result, conditional on φ^* , the engineer's marginal utility from increasing φ becomes constant from her perspective. From (3.13), this marginal utility is positive if and only if

$$\varphi^* \leq \varphi_M \equiv \frac{b\sigma_\delta\sigma_u}{2(bc_f + c_e)}.\tag{3.14}$$

The subscript M indicates the threshold of φ^* at which the engineer's marginal utility becomes zero. If $\varphi^* < \varphi_M$, the price impact is sufficiently weak that a marginal increase

in φ raises the engineer's expected utility. If $\varphi^* > \varphi_M$, the price impact becomes strong enough that the resulting increase in expected trading profits is outweighed by the additional technology-related costs, yielding a negative marginal utility.

Engineer's participation condition. For $\varphi^* \leq \varphi_M$, the engineer's marginal utility of φ is nonnegative, and thus (3.13) implies that her participation condition $\bar{U}_e \geq 0$ is satisfied for all $\varphi \in [0, 1]$. When $\varphi^* > \varphi_M$, however, the engineer's marginal utility is negative, so her expected utility may become negative for sufficiently large φ . Indeed, using (3.13), the engineer's participation condition is expressed as an upper bound on φ :

$$\varphi \leq \Omega(\varphi^*) \equiv \sqrt{\frac{2a\varphi^*}{2(bc_f + c_e)\varphi^* - b\sigma_\delta\sigma_u}}. \quad (3.15)$$

$\Omega(\varphi^*)$ is decreasing in φ^* , as a higher equilibrium technology level increases the price impact and makes it more difficult for the engineer to sustain nonnegative expected utility.

Engineer's optimal technology choice. The engineer's optimal technology choice depends on the value of φ^* . First, if $\varphi^* < \varphi_M$, her marginal utility from increasing φ is positive, and she optimally chooses the maximal feasible technology level, $\varphi = 1$. Second, if $\varphi^* = \varphi_M$, her marginal utility is zero, so her expected utility is constant in φ . In this case, any technology level satisfying the firm's retention condition is optimal. Third, if $\varphi^* > \varphi_M$, her marginal utility from increasing φ is negative. Even in this case, the engineer may optimally choose $\varphi > 0$ to secure employment with the firm, provided that employment yields nonnegative utility. Specifically, if the base salary $a > 0$ (the first term on the RHS of (3.13)) is large enough to more than offset the utility loss associated with $\varphi > 0$ (the second term on the RHS of (3.13)), the engineer chooses a positive technology level solely to secure employment; otherwise, she chooses $\varphi = 0$ and remains unemployed. Formally, the engineer chooses the smallest technology level that secures employment, $\varphi = R(\varphi^*)$, whenever there exists a φ satisfying both the retention condition (3.12) and the engineer's participation condition

(3.15). This is the case if and only if $R(\varphi^*) \leq \Omega(\varphi^*)$, or equivalently,

$$\varphi^* \leq \varphi_N \equiv \frac{\sigma_\delta \sigma_u}{2(c_f + c_e)}. \quad (3.16)$$

If instead $\varphi^* > \varphi_N$, no feasible φ satisfies both (3.12) and (3.15), and she optimally chooses $\varphi = 0$. Combining all of the above cases yields the following lemma.

Lemma 3.1. *The engineer's optimal choice of φ as a best response to φ^* is given by*

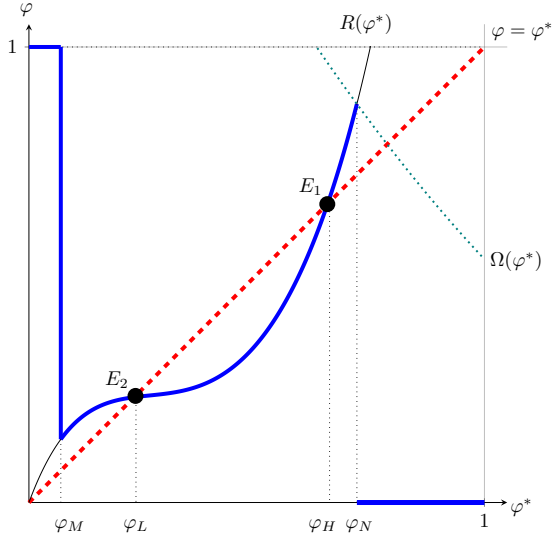
$$\varphi = B(\varphi^*) \equiv \begin{cases} 1 & \text{if } \varphi^* < \varphi_M, \\ \in [R(\varphi^*), 1] & \text{if } \varphi^* = \varphi_M, \\ R(\varphi^*) & \text{if } \varphi_M < \varphi^* \leq \varphi_N, \\ 0 & \text{if } \varphi_N < \varphi^*. \end{cases} \quad (3.17)$$

In the second case, the engineer is indifferent among all technology levels that satisfy the firm's retention condition.

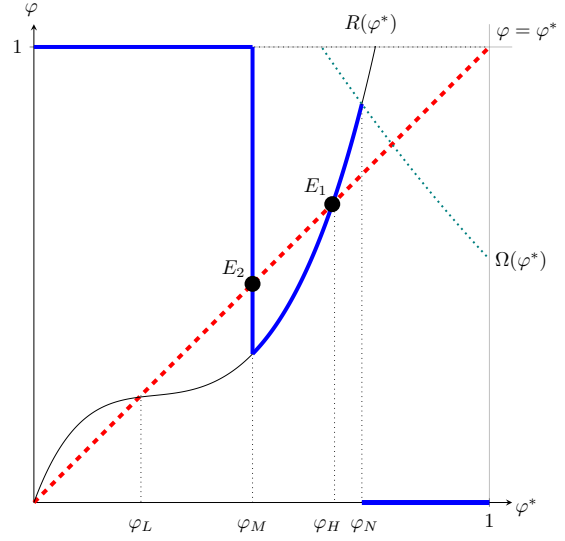
3.4 Equilibrium

In the equilibrium, the engineer's choice of the technology level (3.17) must be consistent with the market maker's belief about its equilibrium level, imposing $\varphi = \varphi^*$. Therefore, the equilibrium is determined by the solutions to the fixed point problem $\varphi^* = B(\varphi^*)$. Figure 2 visualizes this problem by focusing on the parameter values that support the existence of equilibria: the blue solid curve depicts $B(\varphi^*)$, and the red dashed (45-degree) line suggests the belief-consistency condition in the equilibrium.

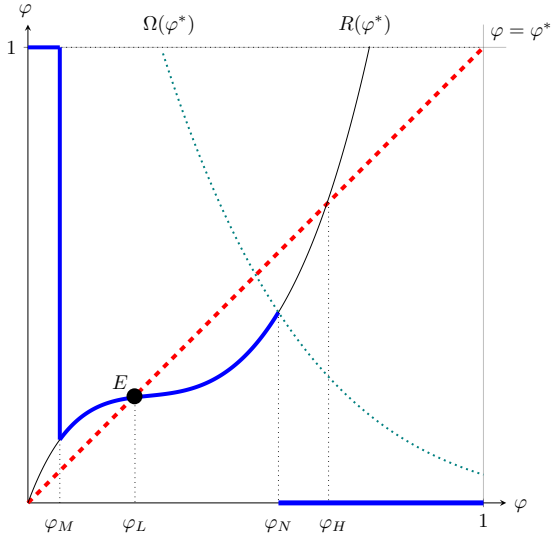
The key feature of the engineer's best response, $\varphi = B(\varphi^*)$, is the upward-sloping curve arising from the binding retention condition, i.e., $\varphi = R(\varphi^*)$ with $\frac{dR(\varphi^*)}{d\varphi^*} > 0$. This represents the strategic complementarity between the engineer's technology development (φ) and its equilibrium level anticipated by the market maker (φ^*). Intuitively, when φ^* is high, the



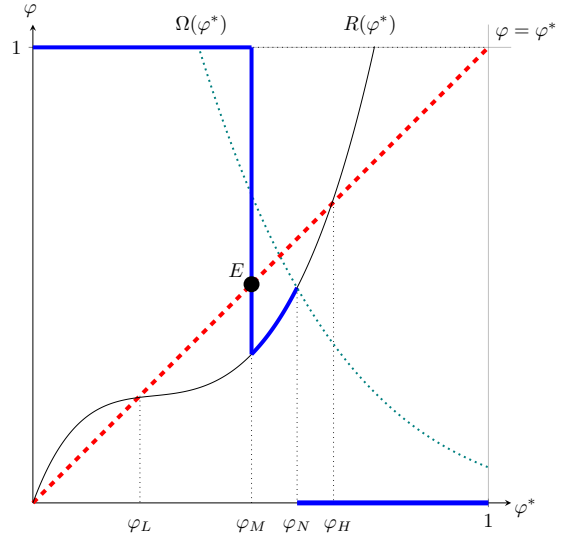
(i): Multiple equilibria at φ_L & φ_H



(ii): Multiple equilibria at φ_M & φ_H



(iii): Unique equilibrium at φ_L



(iv): Unique equilibrium at φ_M

Figure 2: Equilibrium Technology Level

Note: The blue solid lines represent the best-response technology level by the engineer $\varphi = B(\varphi^*)$ given by equation (3.17). The belief-consistency condition, $\varphi = \varphi^*$, is depicted by the red dashed lines. Each black dot represents equilibrium. The numbers of panels correspond to the cases in Proposition 3.1.

order flow is highly informative and thus incurs a strong price impact. In such a market environment, the firm faces a small expected trading profit and is discouraged to keep employing the engineer, as captured by an increase in $R(\varphi^*)$. To meet the retention condition, the engineer responds to high φ^* by actually developing high-level technology, thereby restoring the firm's incentive to retain the engineer. Due to this strategic complementarity, there are multiple technology levels that are consistent with the binding retention condition. Namely, $\varphi^* = R(\varphi^*)$ holds at the following technology levels:

$$\begin{aligned}\varphi_L &\equiv \frac{\sigma_u \sigma_\delta}{4c_f} \left(1 - \sqrt{1 - \frac{16c_f a}{\sigma_u^2 \sigma_\delta^2 (1-b)}} \right), \\ \varphi_H &\equiv \frac{\sigma_u \sigma_\delta}{4c_f} \left(1 + \sqrt{1 - \frac{16c_f a}{\sigma_u^2 \sigma_\delta^2 (1-b)}} \right).\end{aligned}\tag{3.18}$$

Proposition 3.1 below provides analytical characterizations of equilibria using model parameters, where the cases (i)–(iv) correspond to the panels in Figure 2. Also, Figure 3 follows Proposition 3.1 and separates the parameter space according to equilibrium types.

Proposition 3.1. *Define the following cutoffs of the base salary:*

$$a_1 \equiv \frac{(1-b)c_e \sigma_\delta^2 \sigma_u^2}{4(c_f + c_e)^2}\tag{3.19}$$

and

$$a_2 \equiv \frac{b(1-b)c_e \sigma_\delta^2 \sigma_u^2}{4(bc_f + c_e)^2}.\tag{3.20}$$

There are five equilibrium cases depending on the parameter values:

- (i) *If $a \in [\max\{a_1, a_2\}, \hat{a}]$ and $b \in [0, \frac{c_e}{c_f}]$, there are multiple equilibria with self-fulfilling beliefs, one with $\varphi^* = \varphi_L$ and the other with $\varphi^* = \varphi_H$.*
- (ii) *If $a \in [a_1, a_2]$, there are multiple equilibria with self-fulfilling beliefs, one with $\varphi^* = \varphi_M$ and the other with $\varphi^* = \varphi_H$.*
- (iii) *If $a \in [a_2, a_1]$, there is a unique equilibrium with $\varphi^* = \varphi_L$.*

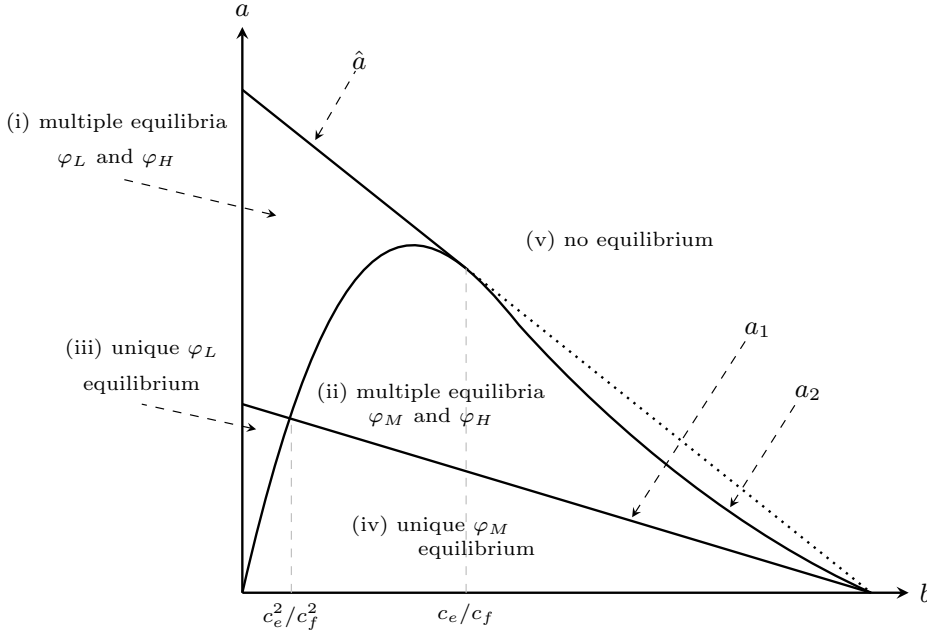


Figure 3: Equilibrium Types and Labor Market Conditions

Note: This figure plots thresholds of the base salary, $\hat{a}$, a_1 , a_2 , that characterize equilibrium types in Proposition 3.1. The number of each region corresponds to that in the proposition.

(iv) If $a \in [0, \min\{a_1, a_2\}]$, there is a unique equilibrium with $\varphi^* = \varphi_M$.

(v) Otherwise, there is no equilibrium.

An important result that emerges from technology development by the engineer is the possibility of multiple equilibria, as shown in (i) and (ii) of Proposition 3.1. Due to the strategic complementarity described above, if the market maker, for whatever reason, believes that $\varphi^* = \varphi_H$ is realized in the equilibrium, it becomes optimal for the engineer to actually choose this high technology level, supporting $\varphi^* = \varphi_H$ as an equilibrium. Potentially, she could deviate from this belief. Nevertheless, she conforms to the market's belief, as the deviation would lead to termination and the loss of her wage income. As the same logic supports an equilibrium with a relatively low-level technology at φ_L or φ_M , multiple self-fulfilling outcomes arise. We refer to the equilibrium with φ_H as the “high-tech” equilibrium, while that with φ_L or φ_M is the “low-tech” equilibrium.¹⁹

¹⁹This classification is based on the equilibrium analysis in the absence of technology opacity in Section 4.1, where we show that the equilibrium is unique and corresponds either to $\varphi^* = \varphi_M$ or to $\varphi^* = \varphi_L$.

Proposition 3.1 attests that multiple equilibria arise when the engineer's base salary is sufficiently high ($a \geq a_1$). From the firm's perspective, a high base salary constitutes a large fixed cost of retention, making the retention condition $\varphi \geq R(\varphi^*)$ in (3.12) more likely to bind. The binding retention condition, in turn, activates the strategic complementarity between φ and φ^* , thereby opening the door to self-fulfilling multiplicity. From the engineer's perspective, although her marginal utility of technology development is negative at $\varphi = \varphi_H$ (since $\varphi_H > \varphi_M$), a sufficiently large base salary provides her with enough fixed rents upon avoiding termination to more than offset the high technology development and maintenance costs. The engineer therefore finds it optimal to deliver the high technology level φ_H merely to conform to the market maker's belief and avoid termination. When a is small, by contrast, the engineer finds it not profitable to develop such a high-level technology, and only the low-tech equilibrium arises.

The performance sensitivity b of compensation plays two related roles. First, holding a fixed, a higher b reduces the firm's share of the net trading profit and thereby tightens the retention condition, so that the economy falls into the region of multiple equilibria as b rises. Second, b determines the type of the low-tech equilibrium, either $\varphi^* = \varphi_L$ or $\varphi^* = \varphi_M$. When b is relatively high, the engineer's wage is closely tied to the trading profit, and the incentives of the firm and the engineer are well aligned, so that the low-tech equilibrium coincides with the engineer's unconstrained optimum ($\varphi^* = \varphi_M$). When b is relatively low, by contrast, the incentive misalignment is severe enough that φ_M falls short of the retention threshold, i.e., $\varphi_M < R(\varphi_M)$; the engineer is then compelled to deliver the minimum technology level that secures employment ($\varphi^* = \varphi_L$).

These forces jointly separate the (a, b) parameter space into the five regions reported in Proposition 3.1 and illustrated in Figure 3. For low values of a , the retention condition is slack and the equilibrium is unique, and b determines the type of the low-tech equilibrium (either φ_L or φ_M). As a rises, the retention condition starts to bind, the strategic complementarity between φ and φ^* kicks in, and the high-tech equilibrium $\varphi^* = \varphi_H$ emerges alongside the low-

tech one, generating belief-driven multiplicity. Finally, when a or b (or both) are excessively large, the firm's break-even condition cannot be met at any technology level and equilibrium ceases to exist.

3.5 Market Quality

To explore implications of multiple equilibria for the financial market, we rely on the standard measures of the financial market quality. Firstly, the price impact, $\lambda = \frac{\sigma_\delta}{2\sigma_u}\varphi^*$, measures market illiquidity. Secondly, the price informativeness is defined as the residual uncertainty in the asset's payoff upon observing the price:

$$\frac{\text{Var}[\delta]}{\text{Var}[\delta|p]} = \frac{2}{2 - \varphi^{*2}}. \quad (3.21)$$

Finally, as the measure of market variations, we compute the price volatility:

$$\text{Var}[p] = \frac{\sigma_\delta^2 \varphi^{*2}}{2}. \quad (3.22)$$

All market quality measures are represented as monotonically increasing functions of the equilibrium technology level φ^* , leading to the following result.

Proposition 3.2. *In the high-tech equilibrium, compared to the low-tech equilibrium, the market is less liquid, and the price is more informative but more volatile.*

The market impact of the high-tech equilibrium is intuitive, as the firm trades more intensively on higher quality information generated through a more sophisticated technology compared to the low-tech equilibrium. The order flow reflects highly precise information about the asset's payoff, and the market maker updates the price substantially, leading to a high price impact. Due to the same logic, the price becomes informative about the asset's payoff. However, the price is more volatile, because both the fundamental information and noise trading are amplified in the price due to the high price impact.

3.6 Inefficiency

3.6.1 Pareto Inefficiency

Building on the discussions surrounding technology investments in real financial markets, a natural question arises: Is the level of investment in the high-tech equilibrium inefficient or excessive, either from a welfare or a firm's perspective? As the market maker breaks even, the trading profit and costs arising from the technology development are split between the trading firm, the engineer, and the noise trader. We examine how this surplus allocation differs between the high-tech and low-tech equilibria.

Firstly, applying the belief-consistency condition $\varphi = \varphi^*$ to (3.10), the firm's expected trading profit reduces to $\bar{\pi} = \frac{\sigma_u \sigma_\delta}{2} \varphi^*$, which becomes highest in the high-tech equilibrium. After netting out the wage payment and maintenance cost, however, the firm's expected utility in (3.11) is

$$\bar{U}_f(\varphi^*) = \begin{cases} 0 & \text{if } \varphi^* = \varphi_L \text{ or } \varphi^* = \varphi_H, \\ a_2 - a & \text{if } \varphi^* = \varphi_M. \end{cases} \quad (3.23)$$

When $\varphi^* = \varphi_L$ or $\varphi^* = \varphi_H$, the retention condition is binding, and the trading firm breaks even in expectation after paying the maintenance cost and the wage. Hence, the firm is indifferent between the high-tech and low-tech equilibria. At $\varphi^* = \varphi_M$, in contrast, the engineer is indifferent between lowering and improving φ after incorporating the development cost, suggesting that the trading firm, without incurring the development cost, earns positive utility. This is captured by the second line of (3.23), where $\varphi^* = \varphi_M$ arises only if $a_2 > a$ (Proposition 3.1). Thus, when the low-tech equilibrium involves φ_M , the firm strictly prefers the low-tech equilibrium to the high-tech equilibrium. In summary, the firm is weakly better off under the low-tech equilibrium.

Similarly, the engineer's expected utility in (3.13) is reduced to

$$\bar{U}_e(\varphi^*) = \begin{cases} \frac{a}{1-b} - c_e \varphi^{*2} & \text{if } \varphi^* = \varphi_L \text{ or } \varphi^* = \varphi_H, \\ a & \text{if } \varphi^* = \varphi_M. \end{cases} \quad (3.24)$$

In the presence of multiple equilibria, we observe $\bar{U}_e(\varphi_H) < \bar{U}_e(\varphi_L)$ when the low-tech equilibrium is realized at $\varphi^* = \varphi_L$, and $\bar{U}_e(\varphi_H) < \bar{U}_e(\varphi_M)$ when $\varphi^* = \varphi_M$. Note that φ_H is too high and induces a negative marginal utility for the engineer. However, if the market maker believes that this technology level is realized in equilibrium, the engineer is compelled to develop that level of technology merely to conform to the market maker's belief, since doing otherwise would result in her dismissal. Hence, the engineer is also better off by switching to the low-tech equilibrium.

The noise trader's utility is defined as the expected trading surplus from executing market order $u \sim N(0, \sigma_u^2)$:

$$\bar{U}_n(\varphi^*) = E[(\delta - p)u] = -\frac{\sigma_u \sigma_\delta}{2} \varphi^*. \quad (3.25)$$

Since φ_H is always larger than both φ_L and φ_M , (3.25) implies that the noise trader's utility is lowest in the high-tech equilibrium. Due to the zero-sum nature of the trading stage, $\bar{U}_n$ represents the direct transfer of the adverse selection cost that the informed trading firm imposes on the market maker, suggesting that the firm earns profits at the expense of the noise trader. As the high-tech equilibrium induces the largest adverse selection cost, the price impact becomes the highest among the three equilibria, and the noise trader experiences the lowest expected utility.

Overall, when the parameters admit multiple equilibria, the engineer and the noise trader are strictly better off if the economy switches from the high-tech equilibrium (φ_H) to the other one (either φ_M or φ_L). As the trading firm's utility either stays unaffected or strictly increases due to this switch, while the market maker is unaffected, we obtain the following result.

Proposition 3.3. *The high-tech equilibrium is Pareto inferior to other equilibria.*

Notably, this Pareto ranking is not an immediate corollary of the decline in total welfare. Even when the total surplus shrinks, the firm could capture a disproportionately large share of the higher trading profits in the high-tech equilibrium and end up strictly better off. Our model shows that this does not occur in equilibrium. Intuitively, the high-tech equilibrium can only be sustained if the firm commits to a sufficiently large base salary a to motivate the engineer’s costly development; this commitment absorbs the additional trading profits and pins the firm’s utility to zero in both equilibria. The engineer thus bears the entire welfare loss: although she receives a higher base wage at the high-tech equilibrium, the development cost grows faster than her compensation, leaving her strictly worse off. As a result, neither the firm nor the engineer benefits from the high-tech equilibrium, while the noise trader unambiguously suffers through the larger price impact.

This result corroborates the idea that excessive investment in financial technology can be socially inefficient (Glode, Green, and Lowery, 2012; Biais, Foucault, and Moinas, 2015; Budish, Cramton, and Shim, 2015; Fishman and Parker, 2015; Bolton, Santos, and Scheinkman, 2016). In this literature, however, inefficiency is defined in terms of aggregate welfare, whereas our model delivers the stronger result that the inefficient outcome is Pareto dominated.²⁰ More importantly, the source of inefficiency differs: in these models, competing traders impose an externality on one another, whereas our inefficiency stems from the bilateral relationship between a firm and its engineer. This difference has a policy implication: contrary to the conventional view, excessive technology investment need not be evidence of an externality among traders. Therefore, policies aimed at curbing competition, or even consolidating trading firms, would not address the source of inefficiency in our setting. Instead, policy discussions should also consider the firm’s internal incentive structure and the opacity of its trading technology.

²⁰Biais, Foucault, and Moinas (2015) obtain a Pareto ranking across multiple equilibria in the case of strategic complementarity, which arises only under a special set of parameter conditions, whereas under their general parameter restrictions, technology-adoption decisions are strategic substitutes and the equilibrium is unique.

3.6.2 Fragility

The overinvestment arising from our mechanism offers unique insights. Namely, the inefficient high-tech equilibrium emerges as a result of self-fulfilling beliefs. Even in the absence of fundamental changes, such as those in the payoff distribution of financial assets or the productivity of financial technologies, a shift in belief alone can trigger an inefficient boom in financial innovation. This result highlights a form of *fragility* in financial technology investment: even minor changes in belief or small perturbations in a parameter can lead to large swings in technology investment and financial market quality. As we discuss below in Section 5, policy interventions in the financial labor market, such as enforcement of non-compete clauses, can push the economy into such regions, unintentionally causing the inefficient outcome.

Furthermore, our result underscores the importance of technological advancements *outside* the financial industry. Innovations in information technologies and AI often originate in non-financial sectors such as the broader tech industry, while these developments tend to trigger major waves of technological adoption and specific-purpose innovations in finance (Jiang, Rebutti, and Zhang, 2025). According to our model, such innovations outside the financial sector can shape the market’s beliefs that the equilibrium technology level is φ_H rather than φ_L or φ_M , which in turn drive excessive technology investments within the financial industry ($\varphi^* = \varphi_H$).

3.7 Empirical Implications

3.7.1 Technology and Financial Market

The high-tech equilibrium in our model appears consistent with real-world phenomena, such as the substantial investment in HFT technologies and, more recently, the increasing interest in applying AI to financial markets. One of the model’s contributions is to provide a formal criterion for empirically assessing whether such investment booms are indeed inefficient.

Proposition 3.4. *The equilibrium technology level φ^* exhibits the following reactions to changes in the contractual conditions:*

- (i) φ_H is monotonically decreasing in the engineer's fixed base salary a , φ_L is monotonically increasing in a , and φ_M is independent of a .
- (ii) φ_H is monotonically decreasing in the performance sensitivity b of the engineer's compensation, while φ_L and φ_M are monotonically increasing in b .
- (iii) The financial market quality measures (price impact, informativeness, and volatility) respond monotonically and in the same direction to changes in φ^* within each equilibrium regime.

Proposition 3.4 provides a way to empirically distinguish between the low-tech and high-tech equilibria. If the equilibrium technology level increases in response to a rise in the engineer's base salary a or the performance sensitivity b of compensation, the economy is in the low-tech equilibrium. In contrast, if the technology level decreases, the economy is in the inefficient high-tech equilibrium characterized by overengineering. When the compensation conditions become more favorable for the engineer, the firm's profit function shifts downward, discouraging the firm from retaining the engineer. To maintain the firm's retention motivation, the engineer needs to adjust the technology development. At the low-tech equilibrium (φ_L or φ_M), the firm's profit curve is increasing in φ due to a relatively low price impact. Thus, the engineer improves the technology level to maintain the firm's utility, leading to increases in the price impact, its volatility, and informativeness. At the high-tech equilibrium, however, φ_H is an excessive investment, and the marginal impact of further technological improvements on the firm's utility is negative. The engineer optimally lowers the technology level in response to higher a or b , leading to the opposite reactions of φ_H and financial market quality measures compared to the low-tech equilibrium.²¹

The values of (a, b) plausibly depend on the labor-market institutions and regulatory envi-

²¹The base salary a becomes irrelevant to the equilibrium technology level when $\varphi^* = \varphi_M$, as a does not affect the engineer's marginal utility of technology development.

ronment, such as the enforceability of non-disclosure and non-compete agreements (Johnson, Lavetti, and Lipsitz, 2023), restrictions on bonus structures, performance-linked pay, or minimum wages (Zeira, 1998; Freixas and Rochet, 2013; Albuquerque, Cabral, and Guedes, 2019), and other rules shaping the design of firm-employee contracts. Our framework offers several implications for how variation in (a, b) interacts with technology investment and financial market outcomes. First, changes in (a, b) can move an economy from a unique equilibrium into a region of multiple equilibria, raising the possibility of Pareto-inefficient overengineering. Second, once multiple equilibria coexist, the impact of such institutional changes on technology investment, price impact, and market liquidity becomes ambiguous, since it depends on which equilibrium the economy is in. This may help reconcile the mixed empirical evidence on the consequences of compensation and contracting reforms (e.g., Saxenian, 1996; Werner, 2023; Jeffers, 2024). Third, although the firm's technology is itself unobservable to the econometrician, a well-identified shock to (a, b) can reveal which equilibrium the economy is in (Proposition 3.4).

3.7.2 Engineer Compensation

Would the engineer's salary also increase when overinvestment occurs? The expected wage transfer in the equilibrium is

$$w(\varphi^*) = \begin{cases} \frac{a}{1-b} & \text{if } \varphi^* = \varphi_L \text{ or } \varphi^* = \varphi_H, \\ a + \frac{b^2 \sigma_a^2 \sigma_\delta^2 c_e}{4(bc_f + c_e)^2} & \text{if } \varphi^* = \varphi_M. \end{cases} \quad (3.26)$$

Regardless of the equilibrium type, $w(\varphi^*)$ is monotonically increasing in the base salary a and the performance sensitivity b . Also, comparing wages across multiple equilibria with parameter values being fixed, $w(\varphi_L) = w(\varphi_H)$ when the low-tech equilibrium involves $\varphi^* = \varphi_L$, and $w(\varphi_M) > w(\varphi_H)$ when it involves $\varphi^* = \varphi_M$.²² These results help analyze two sources

²²Direct comparison implies $w(\varphi_M) - w(\varphi_H) = a_2 - a > 0$, where the inequality holds because multiple equilibria with φ_M and φ_H arise only if $a < a_2$ as shown in Proposition 3.1.

of wage variation: differences across parameter regions and those across multiple equilibria conditional on parameter values.

Firstly, Proposition 3.1 indicates that the economy with a unique low-tech equilibrium leads to modest wages for the engineer, as the equilibrium is unique only when both a and b are relatively low. In contrast, when these parameters increase, not only the engineer compensation grows, but also the economy falls in the regions for multiple equilibria. Therefore, large a and b induce financial engineers to earn higher average wages in the equilibrium with inefficiently large technology investments, compared to economies where only the modest level of investment arises as a unique equilibrium. This result aligns with observations in reality: technology investment booms, such as the recent surge in AI-related investment, coincide with heightened compensation of technical talent (Philippon and Reshef, 2012). In particular, the high specialization and limited substitutability of tech talent may effectively increase the base salary a through, for example, heightened engineers' bargaining power, making a high-wage and high-technology equilibrium more likely to emerge.

Secondly, however, conditional on the economy being in regions for multiple equilibria, and assuming that model parameters are fixed, wages in the high-tech equilibrium are paradoxically lower than in the low-tech one. This counterintuitive result reflects a form of inefficiency: in the high-tech equilibrium, the firm's excessively large maintenance cost $c_f\varphi^2$ crowds out labor compensation. Therefore, our results highlight that the higher equilibrium wages observed during innovation surges are driven primarily by the underlying labor market environment (i.e., the values of a and b) that support substantial innovation rather than by equilibrium selection between high-tech and low-tech outcomes.

4 Sources of Multiple Equilibria

Our analyses so far have established the existence of multiple equilibria and overengineering. A natural question is which features of our framework drive these phenomena. We

identify two essential ingredients that distinguish our setting from the existing literature: (i) technological opacity, which prevents the market maker from observing the engineer's choice, and (ii) engineer-led technology development, in which the entity choosing the technology level (the engineer) is distinct from the one deploying it in the financial market (the firm), creating an incentive problem between the two. In this section, we show that replacing either ingredient with its standard counterpart—transparent technology (Section 4.1) or firm-led development (Section 4.2)—yields a unique equilibrium and eliminates overengineering, demonstrating that both ingredients are jointly necessary for our main findings.

4.1 Transparent Technology

If technology is transparent, the engineer's choice φ is publicly observable, so the market maker's belief must follow φ , imposing $\varphi^* = \varphi$ on and off equilibrium paths. Accordingly, the expected trading profit in equation (3.10) reduces to $\bar{\pi} = \frac{\sigma_u \sigma_\delta}{2} \varphi$, and the expected utilities of the firm and the engineer in (3.11) and (3.13) become

$$\bar{U}_f = (1 - b) \left(\frac{\sigma_\delta \sigma_u}{2} - c_f \varphi \right) \varphi - a, \quad (4.1)$$

and

$$\bar{U}_e = a + \left[\frac{b \sigma_\delta \sigma_u}{2} - (b c_f + c_e) \varphi \right] \varphi. \quad (4.2)$$

From (4.1), the firm's retention condition, $\bar{U}_f \geq 0$, is equivalent to $\varphi_L \leq \varphi \leq \varphi_H$, where φ_L and φ_H are given by (3.18).

Since the engineer's expected utility $\bar{U}_e$ in (4.2) exhibits a single-peaked curve with respect to φ , the equilibrium technology level φ^* is determined either by the engineer's unconstrained optimum level, φ_e (defined below), or by the binding retention condition.²³

²³Condition (ii) in Assumption 2.1 ensures that the engineer's participation constraint is slack at the optimal technology level.

Corollary 4.1. *If technology is transparent, the equilibrium is unique at*

$$\varphi^* = \begin{cases} \varphi_e \equiv \frac{b\sigma_u\sigma_\delta}{4(c_e+bc_f)} & \text{if } a \leq a_0, \\ \varphi_L & \text{if } a > a_0, \end{cases} \quad (4.3)$$

with the threshold of the base salary being

$$a_0 \equiv \frac{\sigma_u^2\sigma_\delta^2b(1-b)(bc_f+2c_e)}{16(bc_f+c_e)^2}. \quad (4.4)$$

The uniqueness of the equilibrium under transparency stems from the absence of strategic interaction between the engineer's technology choice (φ) and the market maker's belief (φ^*). Because the engineer can always influence the market maker's belief and fully internalizes the price impact of φ , the strategic complementarity under opaque technology does not arise.

As in the opaque equilibrium, the retention condition is binding ($\varphi^* = \varphi_L$) when the base wage is relatively high, $a > a_0$.²⁴ While a does not influence the engineer's unconstrained optimal technology level φ_e , it affects her choice through the retention condition: a rise in a makes the condition more stringent, thereby raising the minimum technology level required for retention, φ_L . Consequently, for sufficiently large a , the engineer deviates from her unconstrained optimum φ_e and is forced to choose φ_L instead.

4.2 Firm-led Technology Development

This subsection demonstrates that the separation between technology development and financial decision-making, which creates an incentive problem between them, is critical for the results in Section 3. Overengineering would not arise if the firm were responsible for both technology development and financial investment.

Consider a hypothetical setting in which the firm itself controls φ , while any technology

²⁴The upper threshold of the retention condition, φ_H , does not constrain the engineer's choice, as it is too high from her perspective, and the marginal utility is negative, i.e., $\varphi_e < \varphi_H$ always holds.

development ($\varphi > 0$) requires the engineer's participation, effort input, and wage payment. The engineer accepts this role as long as her expected utility from developing φ is non-negative. Technology is assumed to be opaque, as in Section 3.

The firm chooses φ to maximize $\bar{U}_f$ in equation (3.11) incorporating the engineer's participation condition, $\bar{U}_e \geq 0$. According to Section 3, the engineer is willing to participate and develop any $\varphi \in [0, 1]$ when the market maker's belief is relatively low and the price impact is weak, so that the engineer's marginal utility is positive, as specified by $\varphi^* \leq \varphi_M$ in (3.14). Otherwise, the engineer's participation condition sets the upper bound on the firm's technology choice, $\varphi \leq \Omega(\varphi^*)$, as shown in equation (3.15). Note that the firm may also choose not to produce technology and stay inactive, leading to zero utilities for both the firm and the engineer.

4.2.1 Firm's Optimal Technology Choice

The following lemma provides the best-response technology choice by the firm to the market maker's belief φ^* , denoted as $\varphi = B_f(\varphi^*)$:

Lemma 4.1. *The firm's best response to φ^* is given by*

$$\varphi = B_f(\varphi^*) \equiv \begin{cases} 1 & \text{if } \varphi^* \leq \varphi_\Omega, \\ \Omega(\varphi^*) & \text{if } \varphi_\Omega < \varphi^* \leq \varphi_0, \\ 0 & \text{if } \varphi_0 < \varphi^*, \end{cases} \quad (4.5)$$

where φ_Ω solves $\Omega(\varphi_\Omega) = 1$, and $\varphi_0 = \frac{\sigma_u \sigma_\delta}{2(c_f + c_e)}$.

Figure 4 plots the best-response function in (4.5) against φ^* as a solid line. From (3.11), the firm prefers the highest technology level ($\varphi = 1$) whenever $\varphi^* < \varphi_0$, provided that this choice is feasible. The only constraint that may prevent the firm from doing so is the engineer's participation condition. First, if the market maker's belief is sufficiently low ($\varphi^* \leq \varphi_\Omega$), the engineer is willing to develop $\varphi = 1$, as the relatively low price impact

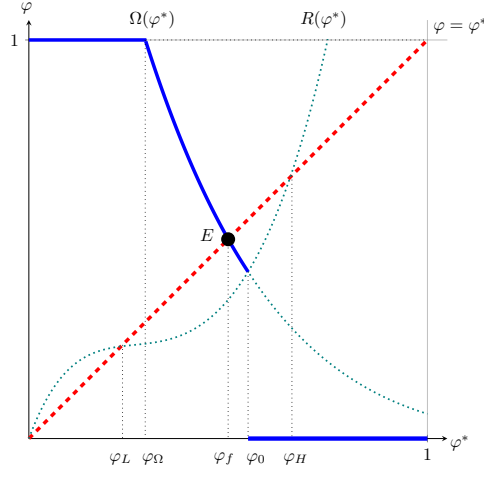


Figure 4: Equilibrium Firm-led Technology Level

ensures a sufficiently large trading profit and positive net utility for the engineer. However, as φ^* increases ($\varphi^* > \varphi_\Omega$), the engineer no longer finds it profitable to develop such high-level technology because the development cost associated with $\varphi = 1$ exceeds her gain from development. Since the firm does not bear this cost and still prefers the highest possible technology level, the engineer's participation condition becomes binding, giving rise to the downward-sloping best-response curve, $B_f(\varphi^*) = \Omega(\varphi^*)$. Finally, as φ^* increases further ($\varphi^* > \varphi_0$), even the highest technology level consistent with the engineer's participation cannot generate enough trading profit to justify the associated technology costs and wage payment. Consequently, the firm opts out of technology development and chooses $\varphi = 0$.

4.3 Equilibrium

Figure 4 illustrates the determination of the equilibrium when the firm has authority in technology development: φ^* is determined by the fixed-point problem $\varphi^* = B_f(\varphi^*)$ of the firm's best-response technology level $\varphi = B_f(\varphi^*)$ and the belief-consistency condition (the dashed 45-degree line).

Proposition 4.1. *Define the threshold of the base salary as $\hat{a}_f \equiv \frac{(1-b)c_e}{4} \left(\frac{\sigma_u \sigma_\delta}{c_e + c_f} \right)^2$. If $a \leq \hat{a}_f$,*

then there is a unique equilibrium with technology level

$$\varphi_f \equiv \frac{b\sigma_u\sigma_\delta}{4(bc_f + c_e)} \left(1 + \sqrt{1 + 16a \frac{bc_f + c_e}{b^2\sigma_u^2\sigma_\delta^2}} \right). \quad (4.6)$$

Otherwise, there is no equilibrium.²⁵

The uniqueness of equilibrium stems from the strategic substitution between the market maker's belief about the firm's technology (φ^*) and the firm's actual technology-development incentive (φ). As φ^* increases, the higher price impact reduces the marginal benefit of increasing φ for both the firm and the engineer. However, due to the cost asymmetry arising from the engineer-specific effort cost of technology development, the engineer's marginal utility decays faster. This incentive misalignment creates the region of intermediate φ^* where the firm is still willing to increase φ while the engineer faces a negative marginal utility. In such a situation, the firm is compelled to choose φ that just satisfies the engineer's participation condition. As the marginal engineer utility is negative in such a region, an increase in the market's belief φ^* induces the firm to reduce φ to maintain the engineer's participation incentive. Consequently, as shown in equation (4.5) and Figure 4, the firm's best-response function becomes downward-sloping against φ^* , which ensures the uniqueness of equilibrium. This result demonstrates that engineer-led technology development is essential for the strategic complementarity and the multiple equilibria featuring overengineering in Section 3. When the firm is responsible for both technology development and financial investment, strategic substitution arises instead, eliminating the possibility of multiple equilibria.

²⁵When the equilibrium exists, $\varphi_L < \varphi_f < \varphi_H$ holds.

5 Extensions

5.1 Oligopolistic Trading Firms

In a more realistic setting with multiple competing firms, does competition mitigate or exacerbate financial overengineering? To address this question, we extend the model to an environment with multiple firms and engineers. The formal model and equilibrium derivations are provided in the [Appendix](#).

5.1.1 Setting

There are $Z \geq 1$ trading firms, each employing one engineer, indexed by $i = 1, \dots, Z$. Within each firm-engineer pair, the wage structure is the same as in the baseline model with common parameters (a, b) across all pairs. By leveraging technology, trading firm i receives a private signal $s_i = \delta + \epsilon_i$ about the asset's payoff with $\epsilon_i \sim N(0, \sigma_i^2)$. Based on s_i , firm i trades in the financial market as an oligopolistic trader.

Technology level and opacity. Following the benchmark model, each firm's technology level is defined as $\varphi_i \equiv \sqrt{\sigma_\delta^2 / (\sigma_\delta^2 + \sigma_i^2)}$. Due to technological opacity, the market maker cannot observe actual technology choices $\boldsymbol{\varphi} \equiv \{\varphi_i\}_{i=1}^Z$. Instead, she rationally anticipates the equilibrium technology levels, denoted by $\boldsymbol{\varphi}^* \equiv \{\varphi_i^*\}_{i=1}^Z$, and sets the asset's price accordingly. We assume that $\boldsymbol{\varphi}^*$ is common knowledge. Also, firm i observes only its own technology level φ_i and cannot observe those of its rivals. As the firm is rational, it shares the market maker's belief about others' technology, $\boldsymbol{\varphi}_{-i}^* \equiv \{\varphi_j^*\}_{j \neq i}$, and determines its trading and retention strategies based on φ_i and $\boldsymbol{\varphi}_{-i}^*$.

Equilibrium. The equilibrium consists of firms' technology levels $\boldsymbol{\varphi}^*$, their retention decisions, trading quantities $\{x_i\}_{i=1}^Z$, and the asset price p , such that for all i : (i) engineer i chooses φ_i to maximize her expected utility, $E[U_{e,i}]$; (ii) firm i chooses whether to retain engineer i and selects its trading strategy to maximize expected utility, $E[U_{f,i}]$; (iii) the market

maker sets the price according to $p = E \left[\delta \left| \sum_{i=1}^Z x_i + u \right| \right]$; and (iv) every agent's beliefs about other agents' behavior are consistent.

5.1.2 Symmetric Equilibrium

We present the results in the symmetric equilibrium, in which all engineers choose the same technology level φ , and all firms make symmetric retention and trading decisions. All players' beliefs about technology levels are also symmetric, so that $\varphi_i^* = \varphi^*$ for all i .

In such an equilibrium, the retention condition for each firm imposes the minimum technology level required to avoid termination:

$$\varphi \geq R(\varphi^*, Z) \equiv \left(\frac{a}{1 - b \sigma_\delta \sigma_u - \sqrt{Z} \varphi^* (2 + (Z - 1) \varphi^{*2}) c_f} \right)^{\frac{1}{2}}. \quad (5.1)$$

R increases with the market's belief φ^* , potentially generating multiple equilibria.

Proposition 5.1. *There exist thresholds of the base salary, $\hat{a}_Z$ and $a_{Z,1} (< \hat{a}_Z)$, both defined in the Appendix.*

- (i) *If $a \in (a_{Z,1}, \hat{a}_Z)$, multiple equilibria exist. One involves either $\varphi^* = \varphi_L$ or $\varphi^* = \varphi_M$, and the other involves $\varphi^* = \varphi_H$, where φ_L and $\varphi_H (> \varphi_L)$ are solutions to the fixed-point problem $\varphi^* = R(\varphi^*, Z)$ with (5.1), and φ_M is the technology level that drives each engineer's marginal utility to zero.*
- (ii) *If $a \leq a_{Z,1}$, a unique equilibrium exists, with either $\varphi^* = \varphi_L$ or $\varphi^* = \varphi_M$.*
- (iii) *If $a \geq \hat{a}_Z$, no equilibrium exists.*

Note that $\hat{a}_Z$ and $a_{Z,1}$ correspond to $\hat{a}$ and a_1 in the baseline model, showing that the intuition in Proposition 3.1 carries over to the oligopolistic-firms setting.²⁶

²⁶ $\hat{a}_Z$ is monotonically decreasing in b , and $a_{Z,1}$ takes a single-peaked curve against b . These thresholds generate a figure similar to Figure 3 in the baseline model.

5.1.3 Financial Market Competition and Overengineering

The minimum technology level required for retention $R(\varphi^*, Z)$ increases with the number of firms Z , suggesting that the retention condition for each firm becomes tighter as competition in the financial market intensifies. Intuitively, as Z increases, the order flow reflects more information about the asset fundamentals from many informed firms, and the price impact becomes larger. Given φ^* and φ , individual expected trading profits deteriorate, and each firm requires a higher technology level to break even.²⁷ The following proposition translates Proposition 5.1 in terms of the number of trading firms:

Proposition 5.2. *There are two thresholds, Z_0 and Z_1 ($1 < Z_0 < Z_1$), both defined in the Appendix.*

(i) *If $a > a_1$, then multiple equilibria arise when $1 \leq Z \leq Z_1$, where a_1 is given by (3.19) in the baseline model.*

(ii) *If $a < a_1$, then multiple equilibria arise when $Z_0 \leq Z \leq Z_1$.*

In both cases, no equilibrium exists when $Z > Z_1$.

Figure 5 illustrates the results stated in Proposition 5.2. The left and right panels correspond to points (i) and (ii) of the proposition, respectively. First, point (i) of Proposition 5.2 demonstrates the robustness of our main results to the case with competing firms. This parameter region nests the multiple equilibria in the baseline model with a monopolistic firm ($Z = 1$), where overengineering may arise under $a > a_1$ (Proposition 3.1). Second, point (ii) highlights an important scenario: even if overengineering does not arise under monopoly ($Z = 1$), it may emerge once the number of firms increases and the market becomes oligopolistic ($Z \geq Z_0$). Intuitively, as Z increases and each firm's retention condition becomes tighter, engineers aim to avoid dismissal and therefore adjust the technology level to boost trading profits. Since φ_H is excessive relative to the firm's profit-maximizing level, engineers must scale back technology development in the high-tech equilibrium to restore

²⁷These results are common in Kyle-type models with multiple informed traders. See, for example, Admati and Pfleiderer (1988) and Holden and Subrahmanyam (1992).

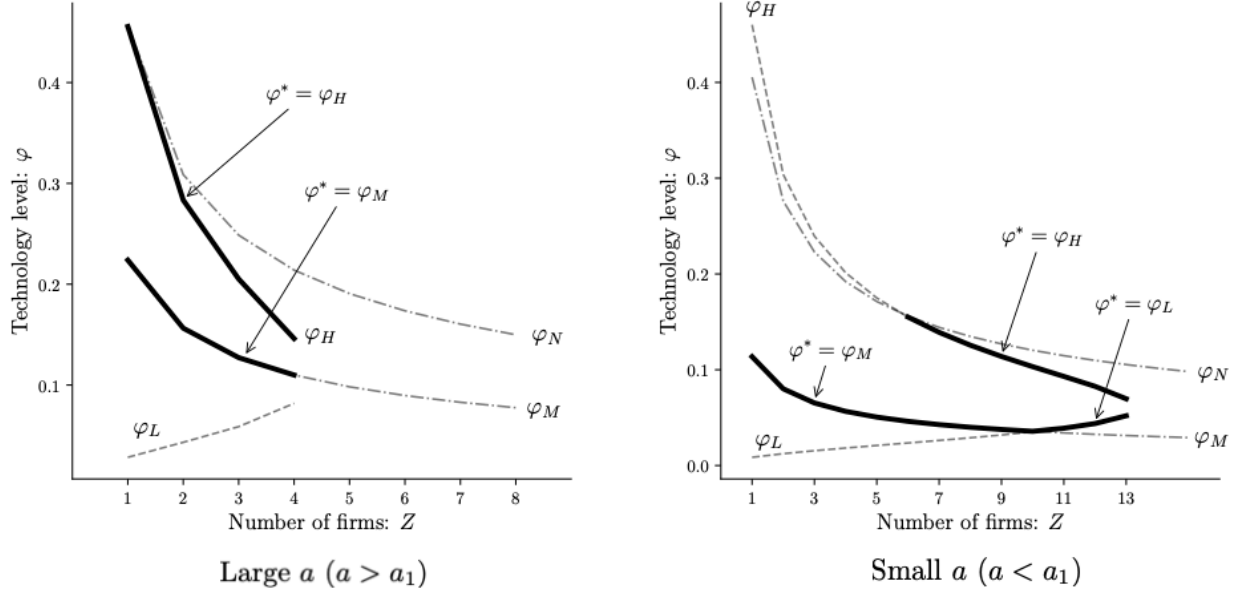


Figure 5: Equilibrium Technology Level in Oligopolistic Firms Model

Note: The figure plots equilibrium technology levels (φ^*) with solid lines, solutions to the binding retention condition (φ_L, φ_H) with dashed lines, and cutoff technology levels (φ_M, φ_N) with dash-dotted lines. Parameter values used in the left panel are $\sigma_\delta = 1$, $\sigma_u = 1.5$, $c_f = 1.6$, $c_e = 0.09$, $b = 0.05$, and $a = 0.02$; those used in the right panel are $\sigma_\delta = 1$, $\sigma_u = 1.5$, $c_f = 1.6$, $c_e = 0.25$, $b = 0.05$, and $a = 0.006$.

the retention condition. This reduction in φ_H , in turn, helps engineers save on technology development costs and, consequently, makes the high-tech equilibrium easier to sustain. This effect is illustrated in the right panel of Figure 5. For small Z , the economy exhibits a unique low-tech equilibrium, whereas for $Z \geq 6$, both the low-tech and high-tech equilibria are sustained. Third, equilibrium disappears when competition among firms becomes too intense ($Z > Z_1$), as it erodes individual trading profits and pushes $R(\varphi^*, Z)$ too high to motivate engineers' technology development.

Moreover, the right panel of Figure 5 shows that the response of the low-tech equilibrium to changes in the number of competing firms can be non-monotonic: φ^* decreases with Z when Z is small, but increases with Z once Z exceeds 10. When the retention condition is slack in the low-tech equilibrium, engineers choose the level of technology such that their marginal utility becomes zero, i.e., $\varphi^* = \varphi_M$. φ_M decreases with Z , reflecting the reduced marginal utility for engineers due to intense competition. Once the retention condition becomes binding, however, $\varphi^* = \varphi_L$ increases with Z , because φ_L falls short of firms' profit-

maximizing level and engineers must develop higher technologies to satisfy the retention requirement.

Overall, this extension highlights the implication of financial market competition for overengineering: technology development tends to be excessive when the financial market is oligopolistic and a small number of major trading firms collectively dominate trading activity. This result is consistent with real-world evidence. For example, [Aquilina, Budish, and O’neill \(2022\)](#) document that latency-arbitrage races by HFTs are concentrated, with the top six firms accounting for over 80% of all race wins and losses. This observation suggests that the most technologically intensive (and often criticized as excessive) trading occur in markets dominated by a limited number of large players.

5.2 Endogenous Opacity

This section is based on the baseline model with a single firm employing one engineer and analyzes the firm’s endogenous choice between transparent and opaque technology. It uncovers the strategic incentives behind opaque innovations and offers new insights, particularly regarding the costs of opacity.

5.2.1 Setup

Before the engineer develops the technology, the trading firm decides on the technology transparency regime, $\chi \in \{0, 1\}$, which is either transparent ($\chi = 0$) or opaque ($\chi = 1$). This choice belongs to the firm rather than the engineer, as it determines how the resulting technology and information will be positioned and protected in operation.

If the firm chooses opaque technology ($\chi = 1$), the equilibrium result presented in Section 3 prevails. If, in contrast, the firm chooses transparent technology ($\chi = 0$), then the result in Subsection 4.1 prevails. These scenarios are referred to as the “opaque equilibrium” and the “transparent equilibrium,” respectively.

To explore the strategic choice by the firm facing multiple equilibria, we introduce an

equilibrium-selection device. In particular, if the parameter values admit multiple equilibria, all players in the model coordinate their beliefs according to a sunspot shock $z \in \{0, 1\}$.²⁸ Namely, with $\theta \equiv \Pr(z = 0)$, no sunspot occurs and the low-tech equilibrium (either φ_M or φ_L) is realized; with the complementary probability $1 - \theta$, a sunspot occurs and the high-tech equilibrium is realized. We assume that the shock is realized after the firm chooses χ but prior to the engineer's technology development.

Equilibrium. The equilibrium in this extended model consists of the baseline equilibrium defined in Definition 1, augmented by the technology's transparency regime, such that the firm chooses $\chi \in \{0, 1\}$ to maximize its *ex-ante* expected utility:

$$\chi^* = \arg \max_{\chi \in \{0, 1\}} \bar{U}_f, \quad (5.2)$$

where $\bar{U}_f$ under the transparent ($\chi = 0$) and opaque ($\chi = 1$) regimes is given by (4.1) and (3.23), respectively, and the expectation accounts for the sunspot shock z .

5.2.2 Equilibrium Opacity

To compare the firm's utility in the transparent and the opaque equilibria, Figure 6 adds the threshold a_0 in Corollary 4.1 to Figure 3. Also Table 1 in Appendix summarizes the technology level and the expected firm utility in both equilibria in each region specified in Figure 6.

Proposition 5.3. *The optimal transparency regime for the firm is characterized as follows:*

- (i) *If $a \in [a_2, \hat{a}]$, then the firm is indifferent between transparent technology ($\chi^* = 0$) and opaque technology ($\chi^* = 1$).*
- (ii) *If $a \leq \min\{a_1, a_2\}$ or $a \in [\max\{a_0, a_1\}, a_2]$, then the firm chooses opaque technology ($\chi^* = 1$).*

²⁸See, for example, Diamond and Dybvig (1983) and Cooper and Ross (1998)

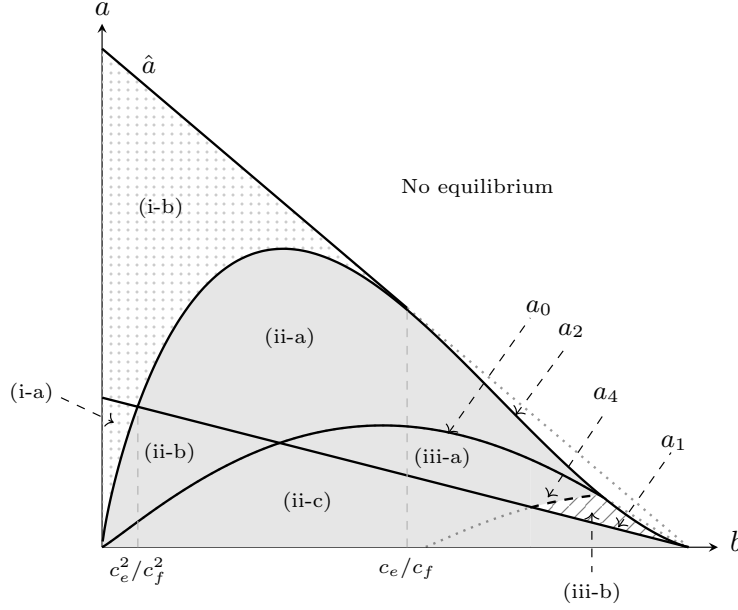


Figure 6: Endogenous Opacity

Note: The figure adds a_0 in equation (4.4) to Figure 3. The dotted areas represent the parameter regions where the firm is indifferent between transparent technology ($\chi = 0$) and opaque technology ($\chi = 1$), the gray areas suggest that $\chi = 1$ is optimal, and the areas with diagonal lines indicate that $\chi = 0$ is optimal. The roman numerals correspond to those in Proposition 5.3.

(iii) If $a \in [a_1, \min\{a_0, \hat{a}\}]$, there is a cutoff of the base salary, $a_4 \equiv \frac{a_0 - \theta a_2}{1 - \theta}$.²⁹ If $a > a_4$, the firm chooses opaque technology ($\chi^* = 1$); otherwise, it chooses transparent technology ($\chi^* = 0$).

Opaque technology leads to the following tradeoff and shapes the equilibrium χ^* . On the one hand, as highlighted in the literature, such as Banerjee and Breon-Drish (2020), Xiong and Yang (2023), and Aoyagi (2025), when technology development is opaque, the price impact is inelastic with respect to φ . As a result, the engineer optimally develops a more advanced technology under opacity than under transparency. In the low-tech equilibrium, where incentive misalignment leads to a technology level that is insufficient from the firm's profit-maximizing perspective, this higher level of technology increases the firm's utility.³⁰

²⁹ a_4 satisfies $a_4 < a_0$ when $b < \frac{2c_e}{c_f}$, while it converges to $a_4 = a_0 = a_2$ at $b = \frac{2c_e}{c_f}$.

³⁰ Under transparency, the individually optimal technology levels for the firm and the engineer are given by $\varphi_f \equiv \frac{\sigma_\delta \sigma_u}{4c_f}$ and $\varphi_e \equiv \frac{b\sigma_\delta \sigma_u}{4(bc_f + c_e)}$, respectively, where the incentive misalignment due to the engineer-specific development cost (c_e) and the hold-up problem due to the performance-based salary ($b < 1$) lead to $\varphi_e < \varphi_f$. Under opacity, the engineer's unconstrained optimal technology level becomes higher than under transparency, i.e., $\varphi_e < \varphi_M$.

On the other hand, opacity may give rise to a self-fulfilling high-tech equilibrium, in which technology development becomes excessive and drives the firm's utility to zero. The emergence of this high-tech equilibrium represents the endogenous cost of opacity and is unique to our model that features belief-driven multiple equilibria.

When the base salary a is small enough that multiple equilibria do not arise under opacity, only the benefit of opacity matters, and developing opaque technology becomes strictly dominant ($\chi^* = 1$). Even when a is large enough that multiplicity emerges, opacity strictly dominates transparency if the performance sensitivity b remains moderate. In this case, the incentive misalignment is sufficiently severe that under transparency the retention condition is binding at $\varphi^* = \varphi_L$ and the firm earns zero utility, while under opacity the low-tech equilibrium φ_M is above the retention requirement and yields strictly positive utility for the firm. The benefit of opacity in the low-tech equilibrium thus outweighs its cost in the high-tech one.

Transparency becomes optimal ($\chi^* = 0$) only when both a is large enough to admit multiplicity under opacity and b is high (region (iii-b) in Figure 6). A high b mitigates the incentive misalignment, since the engineer internalizes a larger share of the trading surplus. Her unconstrained optimum φ_e is already close to the joint-surplus-maximizing level and the retention condition is slack even under transparency. The benefit of opacity is therefore small, while its cost, namely the exposure to the inefficient high-tech equilibrium, remains. Hence the firm strictly prefers transparency.³¹

The mechanism behind the emergence of the transparent equilibrium is unique to our model. The literature focuses on the positive effect of opacity on technology acquisition through an inelastic price impact, which makes the opaque regime optimal for all parameter values.³² Our model instead identifies an endogenous utility cost of opacity arising from

³¹The cutoff, a_4 , shifts upward when θ decreases, expanding the region for $\chi^* = 0$. This is because the high-tech equilibrium becomes more likely, and the cost of opacity grows larger, so that transparent technology becomes more attractive for the firm.

³²To admit transparent information acquisition, the existing models need additional forces, such as competition among informed traders (e.g., [Xiong and Yang, 2023](#)).

belief-driven multiple equilibria and overengineering, allowing both transparent and opaque innovations to coexist in a unified framework.

6 Discussion

6.1 Microfoundations of the Compensation Structure

In the baseline model, we treat the affine wage structure $w = a + b\Pi$ as exogenously given to focus on the equilibrium implications of technological opacity in conjunction with financial markets. To microfound this structure, we modify the setting of the firm's retention decision as follows.

If the firm terminates the engineer, it incurs a salvage cost A and recovers technology with probability $1 - B$, thereby earning the expected trading profit $\bar{\pi}$ in the subsequent trading stage. Intuitively, the technology can be restored, as the firm has observed the details of the developed technology before it fires the engineer, while recovering the technology without the engineer is costly ($A > 0$) and may not succeed ($B > 0$). At the retention decision, the firm's expected utility from dismissing the engineer is given by $(1 - B)(\bar{\pi} - c_f\varphi^2) - A$. If the firm retains the engineer, in contrast, its expected utility after the wage payment is $\bar{\pi} - c_f\varphi^2 - w$.

The wage is determined by Nash bargaining between the firm and the engineer, where we denote the engineer's bargaining power as $\gamma \in (0, 1]$, and the disagreement payoff for the engineer is assumed to be zero.³³ Hence, w is set by maximizing the Nash product: $w^\gamma(B(\bar{\pi} - c_f\varphi^2) - w + A)^{1-\gamma}$, leading to the following result:

³³We normalize the engineer's disagreement payoff to zero. This is consistent with, for example, search frictions and non-disclosure or non-compete obligations that delay her ability to redevelop and redeploy the technology elsewhere, by which point the technology can become obsolete or the asset's fundamental has already been revealed.

Lemma 6.1. *The Nash bargaining solution yields the following wage:*

$$w = \gamma A + \gamma B(\bar{\pi} - c_f \varphi^2), \quad (6.1)$$

which nests the affine wage structure used in the baseline model.

The above setting exactly reproduces the affine wage structure used in the main analysis by denoting $a = \gamma A$ and $b = \gamma B$. Notably, the salvage option strengthens the firm's bargaining position and limits its effective willingness to pay. Hence, even when the engineer has full bargaining power ($\gamma = 1$), she cannot extract the entire surplus, and the incentive problem, which is critical to our main results, persists.

Moreover, Proposition 3.1 reveals that the existence of a strictly positive fixed component in the engineer's compensation ($a > 0$) is the key to the multiple-equilibria result: even when $b = 0$, the region $a \in [a_1, \hat{a}]$ admits both a high-tech and a low-tech equilibrium. The performance sensitivity b enriches these results by, for example, supporting the interior low-tech equilibrium φ_M .³⁴

Beyond the salvage interpretation, the fixed component $a > 0$ can also be conceptually supported by other classical environments, although fully embedding them within our opaque-technology setting is beyond the scope of this paper. A first example is a screening model in the spirit of Rothschild and Stiglitz (1976), in which the firm offers a menu of contracts to engineers of heterogeneous types, leaving the high-type engineer with a constant informational rent.³⁵ A second example is the efficiency-wage framework of Shapiro and Stiglitz (1984), in which a constant wage premium, combined with the threat of dismissal,

³⁴The overengineering equilibrium with φ_H is robust to the introduction of a small ex-ante cost of initiating the project. Although the firm earns zero profits at $\varphi^* = \varphi_L$ and $\varphi^* = \varphi_H$, the interior low-tech equilibrium at $\varphi^* = \varphi_M$ yields strictly positive ex-ante utility, $\theta(a_2 - a) > 0$, and therefore survives arbitrarily small initiation costs.

³⁵Implementing screening in our opaque-technology environment is not analytically tractable: equilibrium multiplicity generates discontinuities in agents' expected utilities at the boundaries of the equilibrium regions, violating the single-crossing and monotonicity properties typically required for screening problems. By contrast, the transparent-technology environment admits a constant informational rent under standard regularity conditions.

disciplines the engineer’s effort and naturally yields a constant component of compensation. A third alternative is a Nash bargaining framework with an exogenous fixed hiring cost for the firm, which generates an affine compensation structure endogenously. These examples collectively suggest that a compensation structure with a positive fixed component emerges naturally from a variety of contracting frictions.

6.2 Acquiring Trading Edges

While our model emphasizes a trading firm’s relationship with a financial engineer who develops proprietary trading technology, the mechanism we identify extends beyond this specific setting. Our framework sheds light on a broader range of circumstances in which firms may inefficiently overinvest in acquiring a trading edge, such as superior information or speed advantages. The key ingredients that drive this inefficiency are (i) information frictions in financial markets (as in [Kyle, 1985](#)), (ii) opacity of the trading edge from the market maker’s perspective, and (iii) the trading edge developed by an entity other than the firm, whose compensation includes a fixed component.

This characterization sharpens the distinction between our setting and existing models of information acquisition. If information is treated as a non-rival good and obtained from a monopolistic seller (e.g., an external analyst as in [Admati and Pfleiderer, 1986, 1988, 1990](#)) or through a perfectly competitive market (as in [Veldkamp, 2006](#)), multiple equilibria do not arise: the information provider either extracts the entire trading surplus or is compensated at the marginal cost of information production, leaving no fixed component anchored on the firm side. By contrast, when a firm hires an in-house analyst who incurs the costs of information discovery and provides it exclusively to the firm, the compensation typically includes a positive fixed component, the firm’s retention condition becomes potentially binding, and overinvestment emerges as a concern. Our framework underscores that overacquisition of a trading edge emerges when it is embedded in firm-specific human capital, hidden from other market participants, and compensated through a contract with a positive fixed component.

This prediction has particular relevance for the recent trend of firms hiring AI talent to develop proprietary trading systems, where opacity, firm-specific human capital, and structured compensation contracts are all inherent features.

7 Conclusion

This paper studies a model of a financial market with information asymmetry, where a trading firm has an engineer on staff, and she develops proprietary technology for an informational advantage in the market. The technology development involves an incentive misalignment between the firm and the engineer, as the engineer alone incurs non-pecuniary effort costs. We show that opaque technology, where the market maker cannot observe the technology level of the trading firm, generates strategic complementarity between the engineer’s innovation incentive and the market maker’s belief about it. Consequently, the model features multiple self-fulfilling equilibria, one of which involves excessive and Pareto-inefficient technology investments. In this “high-tech” equilibrium, the trading firm adopts more aggressive trading strategies, and the price becomes more informative, while it also leads to an illiquid market with a highly volatile price. The high-tech equilibrium exhibits distinctive comparative statics compared to a low-tech equilibrium, providing an empirical tool to identify inefficiency in financial technology investments.

As a direction for future research, it would be interesting to extend our theory of technology investment to a broader growth framework. This could shed light on whether the inefficiencies arising from strategic complementarities and multiple equilibria are unique to the financial sector or also relevant to technology investment in other industries, which could have broader macroeconomic implications.

Appendix

A Proof of Proposition 3.1

Firstly, it holds that $\varphi_M < \varphi_N$ for all parameter values. φ_L and φ_H are the solutions to the binding retention condition in (3.12), which we rewrite as $I(\varphi) = 0$ with

$$I(\varphi) = 2c_f\varphi^2 - \sigma_u\sigma_\delta\varphi + 2\frac{a}{1-b} = 0. \quad (\text{A.1})$$

Firstly, the determinant of $I = 0$ is positive if, and only if,

$$a < \hat{a} \equiv \frac{(1-b)\sigma_\delta^2\sigma_u^2}{16c_f}, \quad (\text{A.2})$$

imposing the first condition in Assumption 2.1. Also,

$$I(1) < 0 \Leftrightarrow a > (1-b) \left(\frac{\sigma_\delta\sigma_u}{2} - c_f \right), \quad (\text{A.3})$$

and

$$I'(1) < 0 \Leftrightarrow c_f > \frac{\sigma_\delta\sigma_u}{4}. \quad (\text{A.4})$$

Hence, these conditions are satisfied if $c_f > \frac{\sigma_\delta\sigma_u}{2}$, and the second condition in Assumption 2.1 ensures $0 < \varphi_L < \varphi_H < 1$.

The tipping point of I is $\varphi = \frac{\sigma_u\sigma_\delta}{4c_f}$, and $\varphi_M \in [\varphi_L, \varphi_H]$ holds if, and only if, $I(\varphi_M) > 0$. This inequality is equivalent to

$$a \leq a_2(b) \equiv \frac{b(1-b)}{4} \frac{c_e\sigma_u^2\sigma_\delta^2}{(bc_f + c_e)^2}. \quad (\text{A.5})$$

Otherwise, $\varphi_M < \varphi_L$ holds when $b < \frac{c_e}{c_f}$, while $\varphi_M > \varphi_H$ holds if $b > \frac{c_e}{c_f}$. Similarly, $\varphi_N \in [\varphi_L, \varphi_H]$ if, and only if, $I(\varphi_N) > 0$, which is equivalent to

$$a \leq a_1(b) \equiv \frac{(1-b)c_e\sigma_u^2\sigma_\delta^2}{(c_e + c_f)^2}. \quad (\text{A.6})$$

Otherwise, due to $c_f > c_e$ in Assumption 2.1, $\varphi_N > \varphi_H$ holds.

a_1 is monotonically decreasing in b with $a_1(1) = 0$, and a_2 is a hump-shaped curve with $a_2(0) = a_2(1) = 0$. Also, $a_1 \geq a_2$ holds for all $b \in [0, 1]$, where a_1 and a_2 are tangent at $b = \frac{c_e}{c_f}$, depicting the curves in Figure 2.

B Proof of Propositions 3.2 and 3.4

Taking the first-order derivative of $\varphi_L, \varphi_H, \varphi_M$, and φ_N with respect to parameters a and b , and applying them to the market quality measures directly lead to the results.

C Proof of Corollary 4.1

As φ_L is the smaller solution to $I(\varphi) = 0$ in equation (A.1), $\varphi_e > \varphi_L$ holds if and only if $I(\varphi_e) > 0$, which leads to $a < a_0$ in equation (4.4).

D Proof of Lemma 4.1 and Proposition 4.1

If $\varphi^* \geq \frac{\sigma_u \sigma_\delta}{2c_f}$, the firm's marginal utility is (weakly) negative and $\varphi = 0$ is optimal. In contrast, if $\varphi^* < \frac{\sigma_u \sigma_\delta}{2c_f}$, then the firm's marginal utility is positive, and the firm potentially seeks to increase φ . Note that, the unconstrained maximum utility for the firm in this scenario is obtained by $\varphi = 1$. At this maximum technology level, $\bar{U}_f = (1-b) \left(\frac{\sigma_u \sigma_\delta}{2\varphi^*} - c_f \right) - a$, and this value is positive if $\varphi^* \leq \varphi_1 \equiv \frac{(1-b)\sigma_u \sigma_\delta}{2(a+(1-b)c_f)}$. Thus, the firm is willing to develop technology if $\varphi^* \leq \varphi_1$. Otherwise, no technology is developed.

Focus on $\varphi^* \leq \varphi_1$. To develop φ , the firm must motivate the engineer's participation, $\bar{U}_e \geq 0$. This participation condition always holds if $\varphi^* \leq \varphi_M$. If, in contrast, $\varphi^* > \varphi_M$, then the firm must follow the engineer's participation condition $\varphi \leq \Omega(\varphi^*)$ in equation (3.15).

Suppose that $a \geq \frac{b}{1-b}c_e$. This inequality is equivalent to $\varphi_1 \leq \varphi_M$: the engineer is willing to participate whenever the firm seeks to increase φ . Hence, the best-response of the firm to φ^* is

$$B_f(\varphi^*) = \begin{cases} 1 & \text{if } \varphi^* < \varphi_1, \\ \in \{0, 1\} & \text{if } \varphi^* = \varphi_1, \\ 0 & \text{if } \varphi^* > \varphi_1, \end{cases} \quad (\text{D.1})$$

where the second line suggests that the firm is indifferent between developing $\varphi = 1$ by hiring the engineer and no technology development. As the equilibrium is determined by the belief consistency condition, $\varphi = \varphi^*$, and the best-response has no intersection with the 45-degree line, there is no equilibrium.

Suppose that $a < \frac{b}{1-b}c_e$ and thus $\varphi_1 > \varphi_M$. Note that $\Omega(\varphi_1) = \sqrt{\frac{a(1-b)}{(1-b)c_e - ab}}$, and $\Omega(\varphi_1) < 1 \Leftrightarrow a < (1-b)c_e$. If $a \geq (1-b)c_e$, then choosing $\varphi = 1$ under $\varphi^* \leq \varphi_1$ always satisfies the engineer's participation constraint. Thus, the best-response function is identical to (D.1), and there is no equilibrium.

If $a < (1 - b)c_e$, then we define $\varphi_\Omega \equiv \frac{b\sigma_u\sigma_\delta}{2(bc_f + c_e - a)}$ so that $\Omega(\varphi_\Omega) = 1$. Note that $\varphi_M < \varphi_\Omega < \varphi_1$ hold. Hence, the engineer's participation condition is slack at $\varphi = 1$ when $\varphi^* \leq \varphi_\Omega$, while it is binding when $\varphi^* > \varphi_\Omega$. In the latter case, setting $\varphi = \Omega(\varphi^*)$ would maximize the firm's utility while ensuring the engineer's participation, which yields

$$\bar{U}_f = (1 - b) \left(\frac{\sigma_u\sigma_\delta}{2\varphi^*} - c_f \right) \frac{a\varphi^*}{\varphi^*(bc_f + c_e) - b\frac{\sigma_u\sigma_\delta}{2}} - a = a \frac{\frac{\sigma_u\sigma_\delta}{2} - \varphi^*(c_f + c_e)}{\varphi^*(bc_f + c_e) - b\frac{\sigma_u\sigma_\delta}{2}}.$$

Note that this is positive if $\varphi^* < \varphi_2 \equiv \frac{\sigma_u\sigma_\delta}{2(c_f + c_e)}$. Also, our condition $a < (1 - b)c_e$ ensures that $\varphi_2 < \varphi_1$. Thus, the best-response of the firm is given by (4.1). Note that the only possibility for the equilibrium existence is that $\Omega(\varphi^*)$ has intersection with $\varphi = \varphi^*$. The fixed-point problem, $\varphi = \Omega(\varphi)$, is rewritten as the following quadratic equation:

$$I_f(\varphi) \equiv \varphi^2(bc_f + c_e) - b\frac{\sigma_u\sigma_\delta}{2}\varphi - a = 0.$$

Thus, a unique equilibrium exists if $\Omega(\varphi_2) \leq \varphi_2$, that is,

$$a < (1 - b)c_e \left(\frac{\sigma_u\sigma_\delta}{2(c_f + c_e)} \right)^2,$$

and the technology level is given by φ_f in Proposition 4.1. Otherwise, there is no equilibrium.

E Proof of Proposition 5.3

Comparing a_0 and a_1 , it holds that

$$a_1 - a_0 = \frac{-(1 - b)\sigma_u^2\sigma_\delta^2c_e^3}{16(c_f + c_e)^2(bc_f + c_e)^2} \Delta a(b), \quad (\text{E.1})$$

where

$$\Delta a(b; l) \equiv (1 - l)^2 lb^2 + 2(1 - l)^2 b - 4, \quad (\text{E.2})$$

with $l \equiv \frac{c_f}{c_e}$. Note that Δa is monotonically increasing in b . Also, $\Delta b(2/l; l) > 0$ when $c_f > 2c_e$ due to the second condition in Assumption 2.1. Hence, $\Delta a = 0$ has a unique solution, given by $b = \frac{2c_e^2}{c_f(c_f - c_e)}$, such that $a_1 < a_0 \Leftrightarrow b > \frac{2c_e^2}{c_f(c_f - c_e)}$. Comparing with a_2 , it holds that

$$a_2 - a_0 = (1 - b)\sigma_u^2\sigma_\delta^2 \frac{2c_f - bc_e}{16(bc_f + c_e)^2}, \quad (\text{E.3})$$

suggesting that $a_2 > a_0 \Leftrightarrow b < \frac{2c_e}{c_f}$.

Finally, a_θ is defined as the threshold for inequality $(a_0 - a) > b\theta(a_2 - a)$, leading to

Region/Regime	Technology quality		Firm's utility	
	Transparent	Opaque	Transparent	Opaque
(i-a)	φ_L	φ_L or φ_H	0	0
(i-b)	φ_L	φ_L	0	0
(ii-a)	φ_L	φ_M or φ_H	0	$\theta(a_2 - a)$
(ii-b)	φ_L	φ_M	0	$a_2 - a$
(ii-c)	φ_e	φ_M	$a_0 - a$	$a_2 - a$
(iii)	φ_e	φ_M or φ_H	$a_0 - a$	$\theta(a_2 - a)$

Table 1: Firm's Expected Utility

Note: This table shows the equilibrium technology level and the firm's expected utility when it chooses transparent ($\chi = 0$) and opaque ($\chi = 1$) technology in each region shown in Figure 6. In the last two columns, the transparency regime that generates higher or equivalent firm utility is highlighted in blue. The table focuses on the parameter values that guarantee the existence of equilibrium in both transparency regimes.

$a_\theta = \frac{a_0 - \theta a_2}{1 - \theta}$. We obtain $a_\theta - a_0 = \frac{\theta}{1 - \theta}(a_0 - a_2)$ and $a_\theta - a_2 = \frac{1}{1 - \theta}(a_0 - a_2)$. Hence, if $b < \frac{2c_e}{c_f}$, then $a_\theta < a_0 < a_2$, while $b > \frac{2c_e}{c_f}$ leads to $a_\theta > a_0 > a_2$. At $b = \frac{2c_e}{c_f}$, these three thresholds intersect with each other.

F Model with Oligopolistic Trading Firms

In this proof, variables with an asterisk (*) represent those defined by equilibrium technology levels, φ^* . Also, for brevity, we omit summation indices where no confusion arises.

F.1 Solution to the Trading Stage

Conditional on the signal, s_i , firm i places a market order x_i and earns trading profit, $\pi_i = (\delta - p)x_i$, where p is set by the market maker based on the aggregate order flow, $y = \sum_i x_i + u$. As in the baseline model, we guess a linear trading strategy $x_i = \beta_i \varphi_i^2 (s_i - \bar{\delta})$ with endogenous trading intensity $\beta_i > 0$.

Market maker's pricing strategy. Under the linear trading strategies, the order flow $y = \sum_i x_i + u$ is informationally equivalent to observing a signal, $s_y = \delta + \epsilon_y$, where the noise

term is defined by $\epsilon_y \equiv \frac{\sum_i \beta_i \varphi_i^2 \epsilon_i + u}{\sum_i \beta_i \varphi_i^2}$ and follows a normal distribution with mean zero and variance

$$\text{Var}[\epsilon_y] = \frac{\sum_i \beta_i^2 \varphi_i^4 \sigma_i^2 + \sigma_u^2}{(\sum_i \beta_i \varphi_i^2)^2}.$$

By solving a standard filtering problem from the market maker's perspective (i.e., following the belief φ^*), the price is set as follows:

$$p = \mathbb{E}[\delta|y] = \frac{\sum_i \beta_i \varphi_i^{*2}}{\sum_i \beta_i^2 \varphi_i^{*2} (1 - \varphi_i^{*2}) + \frac{\sigma_y^2}{\sigma_\delta^2} + (\sum_i \beta_i \varphi_i^{*2})^2} y,$$

suggesting that the price impact is

$$\lambda = \frac{\sum_i \beta_i \varphi_i^{*2}}{\sum_i \beta_i^2 \varphi_i^{*2} (1 - \varphi_i^{*2}) + \frac{\sigma_y^2}{\sigma_\delta^2} + (\sum_i \beta_i \varphi_i^{*2})^2}. \quad (\text{F.1})$$

Firms' trading strategy. Consider firm i . It decides on the trading strategy to maximize the following expected trading profit:

$$\mathbb{E}[\pi_i | s_i] = \left[\left(1 - \lambda \sum_{j \neq i} \beta_j \varphi_j^{*2} \right) \varphi_i^2 s_i - \lambda x_i \right] x_i. \quad (\text{F.2})$$

Note that (F.2) follows the expectation based on firm i 's belief φ_{-i}^* . Taking the first-order condition, the optimal trading quantity is given by

$$x_i = \frac{(1 - \lambda \sum_{j \neq i} \beta_j \varphi_j^{*2}) \varphi_i^2}{2\lambda} s_i,$$

suggesting that the trading intensity is determined as

$$\beta_i = \frac{1}{2\lambda} - \frac{\sum_{j \neq i} \beta_j \varphi_j^{*2}}{2}. \quad (\text{F.3})$$

Equations (F.3) for all i , together with equation (F.1), lead to the following result.

Lemma F.1. *The optimal trading strategy of firm i and the price impact are given by*

$$\beta_i = \frac{\sigma_u}{\sigma_\delta (2 - \varphi_i^{*2})} \frac{1}{\sqrt{\sum_j \frac{\varphi_j^{*2}}{(2 - \varphi_j^{*2})^2}}}, \quad (\text{F.4})$$

and

$$\lambda = \frac{\sigma_\delta}{\sigma_u (1 + \Phi^*)} \sqrt{\sum_j \frac{\varphi_j^{*2}}{(2 - \varphi_j^{*2})^2}}, \quad (\text{F.5})$$

where $\Phi = \sum_i \frac{\varphi_i^2}{2 - \varphi_i^2}$. Firm i 's unconditional expected trading profit is computed as

$$\bar{\pi}_i \equiv \mathbb{E}[\pi_i] = q(\varphi^*) \varphi_i^2 \sigma_\delta^2, \quad (\text{F.6})$$

where

$$q(\varphi^*) \equiv \lambda \beta_i^2 = \frac{\sigma_u}{\sigma_\delta (1 + \Phi^*) (2 - \varphi_i^{*2})^2} \frac{1}{\sqrt{\sum_j \frac{\varphi_j^{*2}}{(2 - \varphi_j^{*2})^2}}}. \quad (\text{F.7})$$

F.2 Firms' Retention Decision

Firm i at the retention stage anticipates that employing engineer i triggers the wage transfer of $w_i = a + b(\bar{\pi}_i - c_f \varphi_i^2)$ to the engineer, so that the retention condition, $\bar{U}_{f,i} \geq 0$, imposes the following lower bound on the technology level:

$$\varphi_i \geq R(\varphi^*, Z) \equiv \sqrt{\frac{a}{(1 - b)(q(\varphi^*) \sigma_\delta^2 - c_f)}}. \quad (\text{F.8})$$

F.3 Engineers' Technology Development

Engineer i 's utility from developing technology is $\bar{U}_{e,i} = a + (bq(\varphi^*) \sigma_\delta^2 - bc_f - c_e) \varphi_i^2$. Firstly, her marginal utility of increasing the technology level is positive if, and only if,

$$q(\varphi^*) > q_M \equiv \frac{bc_f + c_e}{b\sigma_\delta^2}. \quad (\text{F.9})$$

Engineer i chooses the maximum technology level, $\varphi_i = 1$, under (F.9). Secondary, when $q(\varphi^*) < q_M$, the engineer is willing to participate if the following participation condition holds:

$$\varphi_i \leq \Omega(\varphi^*, Z) \equiv \sqrt{\frac{a}{bc_f + c_e - bq(\varphi^*) \sigma_\delta^2}}.$$

Note that no φ_i exists that satisfies both the firm's retention condition and the engineer's participation condition if $\Omega < R$, which is equivalent to the following condition:

$$q(\varphi^*) < q_N \equiv \frac{c_f + c_e}{\sigma_\delta^2}, \quad (\text{F.10})$$

where $q_N < q_M$. In summary, the optimal technology level of engineer i is represented by the following best-response function to φ^* .

$$\varphi_i = B(\varphi^*) = \begin{cases} 1 & \text{if } q_M < q(\varphi^*), \\ \in [R(\varphi^*), 1] & \text{if } q(\varphi^*) = q_M, \\ R(\varphi^*) & \text{if } q_N < q(\varphi^*) < q_M, \\ 0 & \text{if } q(\varphi^*) \leq q_N. \end{cases} \quad (\text{F.11})$$

As in the baseline model, the equilibrium technology levels are determined as solutions to the fixed-point problems involving the best-response functions in (F.11) for all $i = 1, \dots, Z$ and the belief consistency condition, $\varphi = \varphi^*$.

F.4 Proof of Proposition 5.1

Proof. In the symmetric equilibrium, it holds that $\Phi = \frac{Z\varphi^2}{2-\varphi^2}$, and equation (F.7) is reduced to

$$q(\varphi^*) = \frac{\sigma_u}{\sigma_\delta \sqrt{Z} \varphi^* (2 + (Z-1)\varphi^{*2})}. \quad (\text{F.12})$$

Hence, the retention condition (F.8) becomes (5.1). When it is binding, we further rearrange it to obtain the following condition.

$$0 = J(\varphi, Z) \equiv \sqrt{Z} \left(\frac{a}{1-b} + \varphi^2 c_f \right) (2 + (Z-1)\varphi^2) - \varphi \sigma_\delta \sigma_u, \quad (\text{F.13})$$

where function J has the following properties:

$$J_\varphi(\varphi, Z) \equiv \frac{\partial J(\varphi, Z)}{\partial \varphi} = 2\varphi \sqrt{Z} \left[c_f (2 + (Z-1)\varphi^2) + (Z-1) \left(\frac{a}{1-b} + \varphi^2 c_f \right) \right] - \sigma_\delta \sigma_u, \quad (\text{F.14})$$

and $J_{\varphi\varphi} \equiv \frac{\partial^2 J(\varphi, Z)}{\partial \varphi^2} > 0$. Together with $J(0, Z) > 0$, equation (F.13) has at most two solutions, which we denote as φ_L and $\varphi_H (> \varphi_L)$.

First, Assumption 2.1 ensures that $J(1, Z) > 0$ and $J_\varphi(1, Z) > 0$.³⁶ Since $J_\varphi(0, Z) < 0$, there exists a unique solution $\hat{\varphi} \in (0, 1)$ such that $J_\varphi(\hat{\varphi}, Z) = 0$. Since $J(0, Z) > 0$, equation (F.13) has two solutions if, and only if, $J(\hat{\varphi}, Z) < 0$. As $\hat{\varphi}$ is the solution to $J_\varphi = 0$ in (F.14), the envelope condition implies that changes in a influences $J(\hat{\varphi}, Z)$ only through the direct

³⁶ $J(1, Z) > 0$ and $J_\varphi(1, Z) > 0$ are summarized by $\frac{a}{1-b} > \max\left\{ \frac{\sigma_\delta \sigma_u}{\sqrt{Z}(Z+1)} - c_f, \frac{1}{Z-1} \left(\frac{\sigma_\delta \sigma_u}{2\sqrt{Z}} - 2Zc_f \right) \right\}$, while its right-hand side is negative due to Assumption 2.1.

effect. Hence, $J(\hat{\varphi}, Z) < 0$ is rewritten as

$$a < \hat{a}_Z \equiv (1-b)\hat{\varphi} \left(\frac{\sigma_\delta \sigma_u}{\sqrt{Z}(2+(Z-1)\hat{\varphi}^2)} - c_f \hat{\varphi} \right), \quad (\text{F.15})$$

where the inside of the parentheses is positive (as $\hat{\varphi}$ is the solution to $J_\varphi = 0$). Note that $\hat{a}_Z$ is independent of a due to the envelope condition.

Next, consider the engineers' participation condition. In the symmetric equilibrium, conditions (F.9) and (F.10) are rewritten as $q(\varphi^*) > q_j \Leftrightarrow \varphi^* < \varphi_j$ for $j = M$ and N , where φ_M and $\varphi_N (> \varphi_M)$ are, respectively, a unique solution to

$$0 = (Z-1)\varphi^{*3} + 2\varphi^* - \frac{1}{\sqrt{Z}} \frac{\sigma_\delta \sigma_u b}{bc_f + c_e}, \quad (\text{F.16})$$

and

$$0 = (Z-1)\varphi^{*3} + 2\varphi^* - \frac{1}{\sqrt{Z}} \frac{\sigma_\delta \sigma_u}{c_f + c_e}. \quad (\text{F.17})$$

Applying (F.17), it holds that

$$\varphi_N J(\varphi_N, Z) = \left(\frac{a}{1-b} - \varphi_N^2 c_e \right) \frac{\sigma_\delta \sigma_u}{c_f + c_e}. \quad (\text{F.18})$$

Since φ_H is the larger solution to $J = 0$, $\varphi_N > \varphi_H$ holds if $J(\varphi_N, Z) > 0$, that is, $\varphi_N^2 < \frac{1}{c_e} \frac{a}{1-b}$. As φ_N is the solution to (F.17), the condition is equivalent to

$$0 < (Z-1) \left(\frac{1}{c_e} \frac{a}{1-b} \right)^{3/2} + 2 \left(\frac{1}{c_e} \frac{a}{1-b} \right)^{1/2} - \frac{1}{\sqrt{Z}} \frac{\sigma_\delta \sigma_u}{c_f + c_e}. \quad (\text{F.19})$$

The right-hand side is monotonically increasing in ξ , and there is a unique $\xi_{Z,1}$ such that (F.19) holds with equality. Hence, $\varphi_N > \varphi_H$ holds if, and only if, $\xi > \xi_{Z,1}$. In summary, when $\xi_{Z,1} < \xi < \hat{\xi}_Z$, φ_L and φ_H exist, and $\varphi_H < \varphi_N$ holds, suggesting that multiple equilibria exist. $\square$

F.5 Proof of Proposition 5.2

Proof. Observe that φ_H and φ_N are monotonically decreasing in Z , as φ_H is the larger solution to (F.13), and φ_N is the unique solution to (F.17). Moreover, it holds that

$$\left| \frac{d\varphi_H}{dZ} \right| - \left| \frac{d\varphi_N}{dZ} \right| = \frac{a\sigma_\delta \sigma_u}{1-b} \frac{g(\varphi_H) - g(\varphi_N)}{2\varphi_H(\sigma_\delta \sigma_u - lc_f)^2} \frac{d\eta(\varphi_H)}{d\varphi} + g(\varphi_N), \quad (\text{F.20})$$

where $\eta(\varphi, Z) = \sqrt{Z}\varphi(2 + (Z-1)\varphi^2)$, $g(\varphi) \equiv \frac{\varphi(2+(3Z-1)\varphi^2)}{2+3(Z-1)\varphi^2} > 0$, and $\frac{dg}{d\varphi} > 0$. Note also that $\lim_{N \rightarrow \infty} \varphi_N = 0$ and $\varphi_N|_{Z=1} = \frac{\sigma_\delta \sigma_u}{2(c_f + c_e)}$ hold. Therefore, when Z increases, (i) φ_H always stays below φ_N or (ii) φ_H initially lies above φ_N and dips below φ_N at a unique Z_0 . Case (i) arises when $\varphi_H < \varphi_N$ at $Z = 1$. Otherwise, case (ii) arises. When $Z = 1$, the baseline model implies that inequality $\varphi_N < \varphi_H$ holds if and only if $a < a_1$. Thus, if $a > a_1$, then $\varphi_H < \varphi_N$ holds for all Z , i.e., case (ii) arises. Otherwise, $\varphi_H < \varphi_N \Leftrightarrow Z_0 < Z$.

Next, consider the fixed-point problem in equation (F.13). It holds that $\frac{\partial J}{\partial Z} > 0$, and thus Z_1 is defined as the unique Z that solves $J(\hat{\varphi}, Z) = 0$. Such Z_1 uniquely exists, as $\frac{\partial J}{\partial Z} > 0$ and $J_\varphi(\hat{\varphi}, Z) = 0$.

Finally, we show that $Z_0 < Z_1$. It suffices to show that $\varphi_N > \varphi_H$ at $Z = Z_1$. As $J(\hat{\varphi}, Z_1) = 0$, this condition is equivalent to

$$\begin{aligned} 0 &< (Z-1)\hat{\varphi}^3 + 2\hat{\varphi} - \frac{1}{\sqrt{Z}} \frac{\sigma_\delta \sigma_u}{c_f + c_e} \\ &= \frac{c_e - 3c_f}{4\sqrt{Z}c_f(c_f + c_e)} \sigma_\delta \sigma_u - \left(\frac{Z-1}{2c_f} \frac{a}{1-b} - 1 \right) \hat{\varphi}, \end{aligned} \quad (\text{F.21})$$

where the second line comes from $J_\varphi(\hat{\varphi}, Z) = 0$. Since Assumption 2.1 implies $c_f > 2c_e$, the first term of (F.21) is negative. If the coef of the second term is positive, then the proof is completed. If it is negative, then inequality (F.21) is equivalent to

$$\hat{\varphi} < \varphi_0 \equiv \frac{1}{\left(2c_f - (Z-1)\frac{a}{1-b}\right)} \frac{3c_f - c_e}{2\sqrt{Z}(c_f + c_e)} \sigma_\delta \sigma_u.$$

Since $\hat{\varphi}$ is the unique solution to $J_\varphi(\varphi, Z) = 0$, the above inequality is further translated to

$$\sigma_\delta \sigma_u < A \equiv 2\sqrt{Z} \left(2c_f + (Z-1) \frac{\gamma^\xi}{1-\gamma} \right) \varphi_0 + 4\sqrt{Z}c_f(Z-1)\varphi_0^3.$$

By directly applying (F.5),

$$\begin{aligned} A &= \frac{2c_f + (Z-1)\frac{a}{1-b}}{2c_f - (Z-1)\frac{a}{1-b}} \frac{3c_f - c_e}{(c_f + c_e)} \sigma_\delta \sigma_u + 4\sqrt{Z}c_f(Z-1)\varphi_0^3 \\ &> \frac{5}{3} \sigma_\delta \sigma_u, \end{aligned}$$

which confirms that inequality (F.21) always holds and thus concludes the proof. $\square$

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